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Systems and methods for a control station for robotic interventional procedures using a plurality of elongated medical devices

Abstract

A system for controlling a catheter-based procedure system that includes a robotic drive configured to control rotational motion and axial motion of one or more elongated medical devices may include a body, a first control coupled to the body, and a second control coupled to the body. First control is configured to instruct the robotic drive to axially move one of the one or more elongated medical devices in response to manipulation of the first control by a user, and the second control is configured to instruct the robotic drive to rotate one of the one or more elongated medical devices in response to manipulation of the second control by the user, wherein the first control and the second control are positioned on the body so the first control and the second control can be simultaneously manipulated by a first digit and a second digit on a hand of the user.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/597,411, filed Jan. 5, 2022, which is a National Phase application of PCT Application No. PCT/US2020/041985, filed Jul. 14, 2020, which claims priority to U.S. Provisional Patent Application No. 62/874,282, filed Jul. 15, 2019, the contents of which are incorporated by reference herein for all purposes.

FIELD

(1) Embodiments of an input system can be configured to control a catheter-based procedure system in a variety of different modes that use various combinations of input controls as motion and/or selection controls. When used as a motion control, an input control can be configured to control axial and/or rotational movement of at least one elongated medical device. Further, input controls that are used as motion controls may be configured to control movement of an elongated medical device in a position control mode or a speed control mode.

(2) Embodiments relate generally to the field of robotic medical procedure systems and, in particular, to systems, apparatus and methods for robotically controlling the movement and operation of one or more elongated medical devices in robotic interventional procedures.

BACKGROUND

(3) As used herein, the term elongated medical device (EMD) refers to, but is not limited to, catheters (e.g., guide catheters, microcatheters, balloon/stent catheters), wire-based devices (e.g., guidewires, embolization coils, stent retrievers, etc.), and medical devices comprising any combination of these. The term wire-based EMD includes but is not limited to guidewires, microwires, a proximal pusher for embolization coils, stent retrievers, self-expanding stents, and flow divertors. Typically wire-based EMDs do not have a hub or handle at their proximal terminal end.

(4) In one embodiment the EMD is a catheter having a hub at a proximal end of the catheter and a flexible shaft extending from the hub toward the distal end of the catheter, wherein the shaft is more flexible than the hub. In one embodiment the catheter includes an intermediary portion that transitions between the hub and the shaft that has an intermediate flexibility that is less rigid than the hub and more rigid than the shaft. In one embodiment the intermediary portion is a strain relief.

(5) The term drive module refers to the combination of a device module and a cassette.

(6) The term cassette generally refers to the part (non-capital, consumable or sterilizable unit) of the robotic drive system that normally is the sterile interface between a device module and at least one EMD (directly) or through a device adapter (indirectly).

(7) The term device module generally refers to the part (e.g., the capital part) of the robotic drive system that normally contains one or more motors with drive couplers that interface with the cassette.

(8) The term front refers to the side of the robotic drive that faces a bedside user and away from the positioning

system, such as the articulating arm. The term rear refers to the side of the robotic drive that is closest to the positioning system, such as the articulating arm.

(9) The term inwardly refers to the inner portion of a feature. The term outwardly refers to the outer portion of a feature.

(10) The terms top, up, and upper refer to the general direction away from the direction of gravity and the terms bottom, down, and lower refer to the general direction in the direction of gravity.

(11) The terms user or operator refer to a user or operator at a control station. The terms also refer to as a control station user or control station operator.

(12) The terms bedside user or bedside operator refer to a user or operator at a bedside unit.

(13) The term local is used to refer to the location of the patient and bedside unit. For example, a local site is the location of the bedside unit and a patient or subject. At a local site, a user or operator and a control station may be located in the same room or an adjacent room to the patient and bedside unit.

(14) The term remote is used to refer to locations that do not have physical access to the bedside unit and/or patient at a local site. For example, a remote site is a location of a user or operator and a control station used to control the bedside unit remotely. A remote location and a local location are away from one another, for example, in different rooms in the same building, different buildings in the same city, different cities, etc.

(15) The term longitudinal axis of a member (for example, an EMD or other element in the catheter-based procedure system) is the line or axis along the length of the member that passes through the center of the transverse cross section of the member in the direction from a proximal portion of the member to a distal portion of the member. For example, the longitudinal axis of a guidewire is the central axis in the direction from a proximal portion of the guidewire toward a distal portion of the guidewire even though the guidewire may be non-linear in the relevant portion.

(16) The term axial movement of a member refers to translation of the member along the longitudinal axis of the member. For example, when the distal end of an EMD is axially moved in a distal direction along its longitudinal axis into or further into the patient, the EMD is being advanced. When the distal end of an EMD is axially moved in a proximal direction along its longitudinal axis out of or further out of the patient, the EMD is being withdrawn.

(17) The term axial insertion refers to inserting a first member into a second member along the longitudinal axis of the second member. For example, an EMD that is axially loaded in a collet is axially inserted in the collet. An example of axial insertion could be referred to as back loading a catheter on the proximal end of a guidewire.

(18) The term lateral insertion refers to inserting a first member into a second member along a direction in a plane perpendicular to the longitudinal axis of the second member. This can also be referred to as radial loading or side loading.

(19) The term rotational movement of a member refers to the change in angular orientation of the member about the local longitudinal axis of the member. For example, rotational movement of an EMD corresponds to clockwise or counterclockwise rotation of the EMD about its longitudinal axis due to an applied torque.

(20) The term continuous motion refers to motion that does not require a reset and is uninterrupted.

(21) The term discrete motion refers to motion that requires a reset and is interrupted.

(22) The terms distal and proximal define relative locations of two different features. With respect to a robotic drive the terms distal and proximal are defined by the position of the robotic drive in its intended use relative to a patient.

(23) When used to define a relative position, the distal feature is the feature of the robotic drive that is closer to the patient than a proximal feature when the robotic drive is in its intended in-use position. Within a patient, any vasculature landmark further away along the path from the access point is considered more distal than a landmark closer to the access point, where the access point is the point at which the EMD enters the patient.

(24) Similarly, the proximal feature is the feature that is farther from the patient than the distal feature when the robotic drive is in its intended in-use position.

(25) When used to define direction, the distal direction refers to a path on which something is moving or is aimed to move or along which something is pointing or facing from a proximal feature toward a distal feature and/or patient when the robotic drive is in its intended in-use position. The proximal direction is the opposite direction of the distal direction. For example, referring to FIG. 1, a robotic device is shown from the viewpoint of an operator facing a patient. In this arrangement, the distal direction is along the positive X coordinate axis and the proximal direction is along the negative X coordinate axis.

(26) With respect to movement of the individual modules, and referring to FIG. 3, the EMD is moved in a distal direction on a path toward a patient through the introducer interface support 74 which defines the distal end of the robotic drive 24. The proximal end of the robotic drive 24 is the point furthest from the distal end along the negative X axis.

(27) With respect to positions of the individual modules, and referring to FIG. 3, the most distal device module is the device module 32a closest to the distal end of the robotic drive 24. The most proximal device module is the device module 32d positioned furthest from the distal end of the robotic drive 24 along the negative X axis. The relative position of device modules is determined by their relative location to the distal end of the robotic drive. For example, device module 32b is distal to device module 32c.

(28) With respect to distal/proximal portions, sections or ends of an EMD or the robotic drive, and referring to FIG. 3, the portions of cassette **66a** and device module **68a** are defined by their relative location to the distal end of the robotic drive. For example, the distal end of cassette **66a** is the portion of the cassette that is closest to the distal end of the robotic drive and the proximal end of cassette **66a** is the portion of the cassette that is furthest from the distal end of the robotic drive along the negative X axis when the cassette is in-use position on device module **68a**. Stated in another way, the distal end of cassette **66a** is the portion of the cassette through which an EMD is closest to the path leading to a patient in the in-use position.

(29) The term force refers to an agent which causes or tends to cause motion of a body. A force acting on a body may change the motion of the body, retard the motion of the body, balance the forces already acting on the body, and give rise to internal stresses in the body. Characteristics of a force include the magnitude of the force, the line of action of the force (the axis along which the force acts), the direction of the force (corresponding to compressive or tensile force), and the point at which the force is acting.

(30) The term load refers to forces, torques, or combination of forces and torques. The load may include a single component of force (a force along a single axis) or multiple components of forces (multi-axial forces) and/or a single component of torque (a torque around a single axis) or multiple components of torque (multi-axial torque). The load may be static (not change with time) or dynamic (change with time).

(31) The term load sensor refers to a sensor that measures one or more components of force and/or torque. For example, a uniaxial load sensor measures force along one axis or torque about one axis. A multiaxial load sensor measures force and/or torque in multiple mutually orthogonal axes. A load sensor generally generates electrical signals in response to load (for example, a strain gauge based load sensor generates charge in response to load) and generally requires signal conditioning circuitry to convert the signals to force and/or torque. As such, a load sensor is a transducer that converts one or more components of compressive and/or tensile force and/or clockwise and/or counterclockwise torque into a measurable electrical output (for example, voltage or current).

(32) Catheters and other elongated medical devices (EMDs) may be used for minimally invasive medical procedures for the diagnosis and treatment of diseases of various vascular systems, including neurovascular intervention (NVI) also known as neurointerventional surgery, percutaneous coronary intervention (PCI) and peripheral vascular intervention (PVI). These procedures typically involve navigating a guidewire through the vasculature, and via the guidewire advancing a catheter to deliver therapy. The catheterization procedure starts by gaining access into the appropriate vessel, such as an artery or vein, with an introducer sheath using standard percutaneous techniques. Through the introducer sheath, a sheath or guide catheter is then advanced over a diagnostic guidewire to a primary location such as an internal carotid artery for NVI, a coronary ostium for PCI, or a superficial femoral artery for PVI. A guidewire suitable for the vasculature is then navigated through the sheath or guide catheter to a target location in the vasculature. In certain situations, such as in tortuous anatomy, a support catheter or microcatheter is inserted over the guidewire to assist in navigating the guidewire.

(33) The physician or operator may use an imaging system (e.g., fluoroscope) to obtain a cine with a contrast injection and select a fixed frame for use as a roadmap to navigate the guidewire or catheter to the target location, for example, a lesion. Contrast-enhanced images are also obtained while the physician delivers the guidewire or catheter so that the physician can verify that the device is moving along the correct path to the target location. While observing the anatomy using fluoroscopy, the physician manipulates the proximal end of the guidewire or catheter to direct the distal tip into the appropriate vessels toward the lesion or target anatomical location and avoid advancing into side branches.

(34) Robotic catheter-based procedure systems have been developed that may be used to aid a physician in performing catheterization procedures such as, for example, NVI, PCI and PVI. Examples of NVI procedures include coil embolization of aneurysms, liquid embolization of arteriovenous malformations and mechanical thrombectomy of large vessel occlusions in the setting of acute ischemic stroke. In an NVI procedure, the physician uses a robotic system to gain target lesion access by controlling the manipulation of a neurovascular guidewire and microcatheter to deliver the therapy to restore normal blood flow. Target access is enabled by the sheath or guide catheter but may also require an intermediate catheter for more distal territory or to provide adequate support for the microcatheter and guidewire. The distal tip of a guidewire is navigated into, or past, the lesion depending on the type of lesion and treatment. For treating aneurysms, the microcatheter is advanced into the lesion and the guidewire is removed and several embolization coils are deployed into the aneurysm through the microcatheter and used to block blood flow into the aneurysm. For treating arteriovenous malformations, a liquid embolic is injected into the malformation via a microcatheter. Mechanical thrombectomy to treat vessel occlusions can be achieved either through aspiration and/or use of a stent retriever. Depending on the location of the clot, aspiration is either done through an aspiration catheter, or through a microcatheter for smaller arteries. Once the aspiration catheter is at the lesion, negative pressure is applied to remove the clot through the catheter. Alternatively, the clot can be removed by deploying a stent retriever through the microcatheter. Once the clot has integrated into the stent retriever, the clot is retrieved by retracting the stent retriever and microcatheter (or intermediate catheter) into the guide catheter.

(35) In PCI, the physician uses a robotic system to gain lesion access by manipulating a coronary guidewire to deliver

the therapy and restore normal blood flow. The access is enabled by seating a guide catheter in a coronary ostium. The distal tip of the guidewire is navigated past the lesion and, for complex anatomies, a microcatheter may be used to provide adequate support for the guidewire. The blood flow is restored by delivering and deploying a stent or balloon at the lesion. The lesion may need preparation prior to stenting, by either delivering a balloon for pre-dilation of the lesion, or by performing atherectomy using, for example, a laser or rotational atherectomy catheter and a balloon over the guidewire. Diagnostic imaging and physiological measurements may be performed to determine appropriate therapy by using imaging catheters or fractional flow reserve (FFR) measurements.

(36) In PVI, the physician uses a robotic system to deliver the therapy and restore blood flow with techniques similar to NVI. The distal tip of the guidewire is navigated past the lesion and a microcatheter may be used to provide adequate support for the guidewire for complex anatomies. The blood flow is restored by delivering and deploying a stent or balloon to the lesion. As with PCI, lesion preparation and diagnostic imaging may be used as well.

(37) When support at the distal end of a catheter or guidewire is needed, for example, to navigate tortuous or calcified vasculature, to reach distal anatomical locations, or to cross hard lesions, an over-the-wire (OTW) catheter or coaxial system is used. An OTW catheter has a lumen for the guidewire that extends the full length of the catheter. This provides a relatively stable system because the guidewire is supported along the whole length. This system, however, has some disadvantages, including higher friction, and longer overall length compared to rapid-exchange catheters (see below). Typically to remove or exchange an OTW catheter while maintaining the position of the indwelling guidewire, the exposed length (outside of the patient) of guidewire must be longer than the OTW catheter. A 300 cm long guidewire is typically sufficient for this purpose and is often referred to as an exchange length guidewire. Due to the length of the guidewire, two operators are needed to remove or exchange an OTW catheter. This becomes even more challenging if a triple coaxial, known in the art as a tri-axial system, is used (quadruple coaxial catheters have also been known to be used). However, due to its stability, an OTW system is often used in NVI and PVI procedures. On the other hand, PCI procedures often use rapid exchange (or monorail) catheters. The guidewire lumen in a rapid exchange catheter runs only through a distal section of the catheter, called the monorail or rapid exchange (RX) section. With a RX system, the operator manipulates the interventional devices parallel to each other (as opposed to with an OTW system, in which the devices are manipulated in a serial configuration), and the exposed length of guidewire only needs to be slightly longer than the RX section of the catheter. A rapid exchange length guidewire is typically 180-200 cm long. Given the shorter length guidewire and monorail, RX catheters can be exchanged by a single operator. However, RX catheters are often inadequate when more distal support is needed.

(38) When performing vascular interventional procedures, the operator generally uses a set of controls provided at a control station in order to control the robotic system to move each catheter or wire. Each of the controls is typically configured to control a specific device, or to move the catheter or wire in a specific manner. Thus, it is sometimes necessary for the operator to switch between different controls or operate multiple controls simultaneously.

SUMMARY

(39) In accordance with an embodiment, an input system can be configured for controlling a catheter-based procedure system. The catheter-based procedure system can include a robotic drive that may be configured to control rotational motion and axial motion of one or more elongated medical devices. The input system can include a body, a first control, and a second control. The first control can be configured to instruct the robotic drive to axially move one of the one or more elongated medical devices in response to manipulation of the first control by a user, and the second control can be configured to instruct the robotic drive to rotate one of the one or more elongated medical devices in response to manipulation of the second control by the user. The first control and the second control may be positioned on the body so that the first control and the second control can be simultaneously manipulated by a first digit and a second digit on a hand of the user.

(40) In accordance with another embodiment, an input system can be configured for controlling a catheter-based procedure system. The catheter-based procedure system may include a robotic drive that may be configured to control movement of a first elongated medical device and a second elongated medical device. The input system can include a handheld body, a first control, and a second control. The first control can be configured to instruct the robotic drive to move the first elongated medical device in response to manipulation of the first control by a user. The second control can be configured to instruct the robotic drive to move the second elongated medical device in response to manipulation of the second control by the user. Instruction of the robotic drive to move the first elongated medical device may occur simultaneously with instruction of the robotic drive to move the second elongated medical device.

(41) In accordance with another embodiment, an input system can be configured for controlling a catheter-based procedure system. The catheter-based procedure system can include a robotic drive that may be configured to control movement of an elongated medical device. The input system can include a first control and a second control. The first control can be configured to instruct the robotic drive to move the elongated medical device a discrete amount in a first degree of freedom in response to activation of the first control by a user. The second control can be configured to instruct the robotic drive to continuously move the elongated medical device in the first degree of freedom in response to activation of the second control by the user.

(42) In accordance with another embodiment, a method for an input system for controlling a catheter-based procedure

system that includes a robotic drive configured to control rotational motion and axial motion of one or more elongated medical devices includes receiving a first manipulation by a first digit of a first hand of a user of a first control coupled to a body of the input system, receiving a second manipulation by a second digit of the first hand of the user of a second control coupled to the body of the input system, instructing, responsive to the first manipulation, the robotic drive to axially move one of the one or more elongated medical devices, and, responsive to the second manipulation, instructing the robotic drive to rotate one of the one or more elongated medical devices, wherein the first manipulation and the second manipulation occur simultaneously.

(43) In accordance with another embodiment, a method for an input system for controlling a catheter-based procedure system that includes a robotic drive configured to control movement of a first elongated medical device and a second elongated medical device includes receiving a first manipulation of a first control coupled to a handheld body of the input system, receiving a second manipulation of a second control coupled to the handheld body of the input system, instructing, responsive to the first manipulation, the robotic drive to move the first elongated medical device, and, responsive to the second manipulation, instructing the robotic drive to move the second elongated medical device, wherein the first manipulation and the second manipulation occur simultaneously.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments will become more fully understood from the following detailed description, taken in conjunction with the accompanying drawings, wherein the reference numerals refer to like parts in which:
- (2) FIG. 1 is a perspective view of an exemplary catheter-based procedure system in accordance with an embodiment;
- (3) FIG. 2 is a schematic block diagram of an exemplary catheter-based procedure system in accordance with an embodiment;
- (4) FIG. 3 is a perspective view of a robotic drive for a catheter-based procedure system in accordance with an embodiment;
- (5) FIG. 4 is a perspective view of a handheld input system for a catheter-based procedure system in accordance with an embodiment;
- (6) FIG. 5 is a perspective view of a handheld input system including a joystick in accordance with an embodiment;
- (7) FIG. 6 is a detailed perspective view of a joystick in accordance with an embodiment;
- (8) FIG. 7 is a perspective view of a handheld input system including an input control configured as a scroll wheel in accordance with an embodiment;
- (9) FIG. 8 is a detailed perspective view of the scroll wheel of FIG. 7;
- (10) FIG. 9 is a perspective view of a handheld input system including an input control configured as a jog wheel in accordance with an embodiment;
- (11) FIG. 10 is a detailed perspective view of the jog wheel of FIG. 9;
- (12) FIG. 11 is a perspective view of an input control configured as a dial in accordance with an embodiment;
- (13) FIG. 12 is a perspective view of an input control configured as a knob in accordance with an embodiment;
- (14) FIG. 13 is a perspective view of a handheld input system including two scroll wheels in accordance with an embodiment;
- (15) FIG. 13B is a perspective view of a handheld input system including two scroll wheels, a knob and selection buttons in accordance with an embodiment;
- (16) FIG. 14 is a perspective view of a handheld input system including binary input controls, analog input controls, scrolling input controls, and touch input controls in accordance with an embodiment;
- (17) FIG. 15 is a perspective view of an input system including a scroll wheel in accordance with an embodiment;
- (18) FIG. 16 is a perspective view of an input system including two scroll wheels in accordance with an embodiment;
- (19) FIG. 17 is a perspective view of the input system of FIG. 16 with a button pad in accordance with an embodiment;
- (20) FIG. 18 is a perspective view of an input system including two touch input controls in accordance with an embodiment;
- (21) FIG. 19 is a perspective view of a translating input system and a button pad in accordance with an embodiment;
- (22) FIG. 20 is a graphical user interface displaying position information in accordance with an embodiment;
- (23) FIG. 21 is a graphical user interface displaying position information in accordance with an embodiment;
- (24) FIG. 22 is a graphical user interface displaying load information via a linear gauge in accordance with an embodiment;
- (25) FIG. 23 is a graphical user interface displaying load information via a rotary gauge in accordance with an embodiment;
- (26) FIG. 24 is a perspective view of a button pad including binary motion controls in accordance with an embodiment; and
- (27) FIG. 25 is a perspective view of a button pad including scrolling motion controls in accordance with an

embodiment.

DETAILED DESCRIPTION

(28) FIG. 1 is a perspective view of an exemplary catheter-based procedure system **10** in accordance with an embodiment. Catheter-based procedure system **10** may be used to perform catheter-based medical procedures, e.g., percutaneous intervention procedures such as a percutaneous coronary intervention (PCI) (e.g., to treat STEMI), a neurovascular interventional procedure (NVI) (e.g., to treat an emergent large vessel occlusion (ELVO)), peripheral vascular intervention procedures (PVI) (e.g., for critical limb ischemia (CLI), etc.). Catheter-based medical procedures may include diagnostic catheterization procedures during which one or more catheters or other elongated medical devices (EMDs) are used to aid in the diagnosis of a patient's disease. For example, during one embodiment of a catheter-based diagnostic procedure, a contrast media is injected onto one or more arteries through a catheter and an image of the patient's vasculature is taken. Catheter-based medical procedures may also include catheter-based therapeutic procedures (e.g., angioplasty, stent placement, treatment of peripheral vascular disease, clot removal, arterial venous malformation therapy, treatment of aneurysm, etc.) during which a catheter (or other EMD) is used to treat a disease. Therapeutic procedures may be enhanced by the inclusion of adjunct devices **54** (shown in FIG. 2) such as, for example, intravascular ultrasound (IVUS), optical coherence tomography (OCT), fractional flow reserve (FFR), etc. It should be noted, however, that one skilled in the art would recognize that certain specific percutaneous intervention devices or components (e.g., type of guidewire, type of catheter, etc.) may be selected based on the type of procedure that is to be performed. Catheter-based procedure system **10** can perform any number of catheter-based medical procedures with minor adjustments to accommodate the specific percutaneous intervention devices to be used in the procedure.

(29) Catheter-based procedure system **10** includes, among other elements, a bedside unit **20** and a control station **26**. Bedside unit **20** includes a robotic drive **24** and a positioning system **22** that are located adjacent to a patient **12**. Patient **12** is supported on a patient table **18**. The positioning system **22** is used to position and support the robotic drive **24**. The positioning system **22** may be, for example, a robotic arm, an articulated arm, a holder, etc. The positioning system **22** may be attached at one end to, for example, a rail on the patient table **18**, a base, or a cart. The other end of the positioning system **22** is attached to the robotic drive **24**. The positioning system **22** may be moved out of the way (along with the robotic drive **24**) to allow for the patient **12** to be placed on the patient table **18**. Once the patient **12** is positioned on the patient table **18**, the positioning system **22** may be used to situate or position the robotic drive **24** relative to the patient **12** for the procedure. In an embodiment, patient table **18** is operably supported by a pedestal **17**, which is secured to the floor and/or earth. Patient table **18** is able to move with multiple degrees of freedom, for example, roll, pitch, and yaw, relative to the pedestal **17**. Bedside unit **20** may also include controls and displays **46** (shown in FIG. 2). For example, controls and displays may be located on a housing of the robotic drive **24**.

(30) Generally, the robotic drive **24** may be equipped with the appropriate percutaneous interventional devices and accessories **48** (shown in FIG. 2) (e.g., guidewires, various types of catheters including balloon catheters, stent delivery systems, stent retrievers, embolization coils, liquid embolics, aspiration pumps, device to deliver contrast media, medicine, hemostasis valve adapters, syringes, stopcocks, inflation device, etc.) to allow the user or operator **11** to perform a catheter-based medical procedure via a robotic system by operating various controls such as the controls and inputs located at the control station **26**. Bedside unit **20**, and in particular robotic drive **24**, may include any number and/or combination of components to provide bedside unit **20** with the functionality described herein. A user or operator **11** at control station **26** is referred to as the control station user or control station operator and referred to herein as user or operator. A user or operator at bedside unit **20** is referred to as bedside unit user or bedside unit operator. The robotic drive **24** includes a plurality of device modules **32a-d** mounted to a rail or linear member **60** (shown in FIG. 3). The rail or linear member **60** guides and supports the device modules. Each of the device modules **32a-d** may be used to drive an EMD such as a catheter or guidewire. For example, the robotic drive **24** may be used to automatically feed a guidewire into a diagnostic catheter and into a guide catheter in an artery of the patient **12**. One or more devices, such as an EMD, enter the body (e.g., a vessel) of the patient **12** at an insertion point **16** via, for example, an introducer sheath.

(31) Bedside unit **20** is in communication with control station **26**, allowing signals generated by the user inputs of control station **26** to be transmitted wirelessly or via hardwire to bedside unit **20** to control various functions of bedside unit **20**. As discussed below, control station **26** may include a control computing system **34** (shown in FIG. 2) or be coupled to the bedside unit **20** through a control computing system **34**. Bedside unit **20** may also provide feedback signals (e.g., loads, speeds, operating conditions, warning signals, error codes, etc.) to control station **26**, control computing system **34** (shown in FIG. 2), or both.

(32) Communication between the control computing system **34** and various components of the catheter-based procedure system **10** may be provided via a communication link that may be a wireless connection, cable connections, or any other means capable of allowing communication to occur between components. Control station **26** or other similar control system may be located either at a local site (e.g., local control station **38** shown in FIG. 2) or at a remote site (e.g., remote control station and computer system **42** shown in FIG. 2).

(33) Catheter procedure system **10** may be operated by a control station at the local site, a control station at a remote site, or both the local control station and the remote control station at the same time. At a local site, user or operator **11** and control station **26** are located in the same room or an adjacent room to the patient **12** and bedside unit **20**. As used herein, a local site is the location of the bedside unit **20** and a patient **12** or subject (e.g., animal or cadaver) and the remote site is the location of a user or operator **11** and a control station **26** used to control the bedside unit **20** remotely. A control station **26** (and a control computing system) at a remote site and the bedside unit **20** and/or a control computing system at a local site may be in communication using communication systems and services **36** (shown in FIG. 2), for example, through the Internet. In an embodiment, the remote site and the local (patient) site are away from one another, for example, in different rooms in the same building, different buildings in the same city, different cities, or other different locations where the remote site does not have physical access to the bedside unit **20** and/or patient **12** at the local site.

(34) Control station **26** generally includes one or more input systems **28** configured to receive user inputs to operate various components or systems of catheter-based procedure system **10**. In the embodiment shown, control station **26** allows the user or operator **11** to control bedside unit **20** to perform a catheter-based medical procedure. For example, input systems **28** may be configured to cause bedside unit **20** to perform various tasks using percutaneous intervention devices (e.g., EMDs) interfaced with the robotic drive **24** (e.g., to advance, retract, or rotate a guidewire, advance, retract or rotate a catheter, inflate or deflate a balloon located on a catheter, position and/or deploy a stent, position and/or deploy a stent retriever, position and/or deploy a coil, inject contrast media into a catheter, inject liquid embolics into a catheter, inject medicine or saline into a catheter, aspirate on a catheter, or to perform any other function that may be performed as part of a catheter-based medical procedure). Robotic drive **24** includes various drive mechanisms to cause movement (e.g., axial and rotational movement) of the components of the bedside unit **20** including the percutaneous intervention devices.

(35) In one embodiment, input systems **28** may include one or more touch screens, joysticks, scroll wheels, and/or buttons. In addition to input systems **28**, the control station **26** may use additional user controls **44** (shown in FIG. 2) such as foot switches and microphones for voice commands, etc. Input systems **28** may be configured to advance, retract, or rotate various components and percutaneous intervention devices such as, for example, a guidewire, and one or more catheters or microcatheters. Buttons may include, for example, an emergency stop button, a multiplier button, device selection buttons and automated move buttons. When an emergency stop button is pushed, the power (e.g., electrical power) is shut off or removed to bedside unit **20**. When in a speed control mode, a multiplier button acts to increase or decrease the speed at which the associated component is moved in response to a manipulation of input modules **28**. When in a position control mode, a multiplier button changes the mapping between input distance and the output commanded distance.

(36) Device selection buttons allow the user or operator **11** to select which of the percutaneous intervention devices loaded into the robotic drive **24** are controlled by input systems **28**. Automated move buttons are used to enable algorithmic movements that the catheter-based procedure system **10** may perform on a percutaneous intervention device without direct command from the user or operator **11**. In one embodiment, input systems **28** may include one or more controls or icons (not shown) displayed on a touch screen (that may or may not be part of a display **30**), that, when activated, causes operation of a component of the catheter-based procedure system **10**. Input systems **28** may also include a balloon or stent control that is configured to inflate or deflate a balloon and/or deploy a stent. Each of the input systems **28** may include one or more buttons, scroll wheels, joysticks, touch screen, etc. that may be used to control the particular component or components to which the control is dedicated. In addition, one or more touch screens may display one or more icons (not shown) related to various portions of input systems **28** or to various components of catheter-based procedure system **10**.

(37) Control station **26** may include a display **30**. In other embodiments, the control station **26** may include two or more displays **30**. Display **30** may be configured to display information or patient specific data to the user or operator **11** located at control station **26**. For example, display **30** may be configured to display image data (e.g., X-ray images, MRI images, CT images, ultrasound images, etc.), hemodynamic data (e.g., blood pressure, heart rate, etc.), patient record information (e.g., medical history, age, weight, etc.), lesion or treatment assessment data (e.g., IVUS, OCT, FFR, etc.). In addition, display **30** may be configured to display procedure specific information (e.g., procedural checklist, recommendations, duration of procedure, catheter or guidewire position, volume of medicine or contrast agent delivered, etc.). Further, display **30** may be configured to display information to provide the functionalities associated with control computing system **34** (shown in FIG. 2). Display **30** may include touch screen capabilities to provide some of the user input capabilities of the system.

(38) Catheter-based procedure system **10** also includes an imaging system **14**. Imaging system **14** may be any medical imaging system that may be used in conjunction with a catheter based medical procedure (e.g., non-digital X-ray, digital X-ray, CT, MRI, ultrasound, etc.). In an exemplary embodiment, imaging system **14** is a digital X-ray imaging device that is in communication with control station **26**. In one embodiment, imaging system **14** may include a C-arm (shown in FIG. 1) that allows imaging system **14** to partially or completely rotate around patient **12** in order to obtain images at different angular positions relative to patient **12** (e.g., sagittal views, caudal views, anterior-posterior views,

etc.). In one embodiment imaging system **14** is a fluoroscopy system including a C-arm having an X-ray source **13** and a detector **15**, also known as an image intensifier.

(39) Imaging system **14** may be configured to take X-ray images of the appropriate area of patient **12** during a procedure. For example, imaging system **14** may be configured to take one or more X-ray images of the head to diagnose a neurovascular condition. Imaging system **14** may also be configured to take one or more X-ray images (e.g., real time images) during a catheter-based medical procedure to assist the user or operator **11** of control station **26** to properly position a guidewire, guide catheter, microcatheter, stent retriever, coil, stent, balloon, etc. during the procedure. The image or images may be displayed on display **30**. For example, images may be displayed on display **30** to allow the user or operator **11** to accurately move a guide catheter or guidewire into the proper position.

(40) In order to clarify directions, a rectangular coordinate system is introduced with X, Y, and Z axes. The positive X axis is oriented in a longitudinal (axial) distal direction, that is, in the direction from the proximal end to the distal end, stated another way from the proximal to distal direction. The Y and Z axes are in a transverse plane to the X axis, with the positive Z axis oriented up, that is, in the direction opposite of gravity, and the Y axis is automatically determined by right-hand rule.

(41) FIG. **2** is a block diagram of catheter-based procedure system **10** in accordance with an exemplary embodiment. Catheter-procedure system **10** may include a control computing system **34**. Control computing system **34** may physically be, for example, part of control station **26** (shown in FIG. **1**). Control computing system **34** may generally be an electronic control unit suitable to provide catheter-based procedure system **10** with the various functionalities described herein. For example, control computing system **34** may be an embedded system, a dedicated circuit, a general-purpose system programmed with the functionality described herein, etc. Control computing system **34** is in communication with bedside unit **20**, communications systems and services **36** (e.g., Internet, firewalls, cloud services, session managers, a hospital network, etc.), a local control station **38**, additional communications systems **40** (e.g., a telepresence system), a remote control station and computing system **42**, and patient sensors **56** (e.g., electrocardiogram (ECG) devices, electroencephalogram (EEG) devices, blood pressure monitors, temperature monitors, heart rate monitors, respiratory monitors, etc.). The control computing system is also in communication with imaging system **14**, patient table **18**, additional medical systems **50**, contrast injection systems **52** and adjunct devices **54** (e.g., IVUS, OCT, FFR, etc.). The bedside unit **20** includes a robotic drive **24**, a positioning system **22** and may include additional controls and displays **46**. As mentioned above, the additional controls and displays may be located on a housing of the robotic drive **24**. Interventional devices and accessories **48** (e.g., guidewires, catheters, etc.) interface to the bedside system **20**. In an embodiment, interventional devices and accessories **48** may include specialized devices (e.g., IVUS catheter, OCT catheter, FFR wire, diagnostic catheter for contrast, etc.) which interface to their respective adjunct devices **54**, namely, an IVUS system, an OCT system, and FFR system, etc.

(42) In various embodiments, control computing system **34** is configured to generate control signals based on the user's interaction with input systems **28** (e.g., of a control station **26** (shown in FIG. **1**) such as a local control station **38** or a remote control station **42**) and/or based on information accessible to control computing system **34** such that a medical procedure may be performed using catheter-based procedure system **10**. The local control station **38** includes one or more displays **30**, one or more input systems **28**, and additional user controls **44**. The remote control station and computing system **42** may include similar components to the local control station **38**. The remote **42** and local **38** control stations can be different and tailored based on their required functionalities. The additional user controls **44** may include, for example, one or more foot input controls. The foot input control may be configured to allow the user to select functions of the imaging system **14** such as turning on and off the X-ray and scrolling through different stored images. In another embodiment, a foot input device may be configured to allow the user to select which devices are mapped to scroll wheels included in input systems **28**. Additional communication systems **40** (e.g., audio conference, video conference, telepresence, etc.) may be employed to help the operator interact with the patient, medical staff (e.g., angio-suite staff), and/or equipment in the vicinity of the bedside.

(43) Catheter-based procedure system **10** may be connected or configured to include any other systems and/or devices not explicitly shown. For example, catheter-based procedure system **10** may include image processing engines, data storage and archive systems, automatic balloon and/or stent inflation systems, medicine injection systems, medicine tracking and/or logging systems, user logs, encryption systems, systems to restrict access or use of catheter-based procedure system **10**, etc.

(44) As mentioned, control computing system **34** is in communication with bedside unit **20** which includes a robotic drive **24**, a positioning system **22** and may include additional controls and displays **46**, and may provide control signals to the bedside unit **20** to control the operation of the motors and drive mechanisms used to drive the percutaneous intervention devices (e.g., guidewire, catheter, etc.). The various drive mechanisms may be provided as part of a robotic drive **24**. FIG. **3** is a perspective view of a robotic drive for a catheter-based procedure system **10** in accordance with an embodiment. In FIG. **3**, a robotic drive **24** includes multiple device modules **32a-d** coupled to a linear member **60**. Each device module **32a-d** is coupled to the linear member **60** via a stage **62a-d** moveably mounted to the linear member **60**. A device module **32a-d** may be connected to a stage **62a-d** using a connector such as an offset bracket **78a-d**. In another embodiment, the device module **32a-d** is directly mounted to the stage **62a-d**.

Each stage **62a-d** may be independently actuated to move linearly along the linear member **60**. Accordingly, each stage **62a-d** (and the corresponding device module **32a-d** coupled to the stage **62a-d**) may independently move relative to each other and the linear member **60**. A drive mechanism is used to actuate each stage **62a-d**. In the embodiment shown in FIG. 3, the drive mechanism includes independent stage translation motors **64a-d** coupled to each stage **62a-d** and a stage drive mechanism **76**, for example, a lead screw via a rotating nut, a rack via a pinion, a belt via a pinion or pulley, a chain via a sprocket, or the stage translation motors **64a-d** may be linear motors themselves. In some embodiments, the stage drive mechanism **76** may be a combination of these mechanisms, for example, each stage **62a-d** could employ a different type of stage drive mechanism. In an embodiment where the stage drive mechanism is a lead screw and rotating nut, the lead screw may be rotated and each stage **62a-d** may engage and disengage from the lead screw to move, e.g., to advance or retract. In the embodiment shown in FIG. 3, the stages **62a-d** and device modules **32a-d** are in a serial drive configuration.

(45) Each device module **32a-d** includes a device module **68a-d** and a cassette **66a-d** mounted on and coupled to the device module **68a-d**. In the embodiment shown in FIG. 3, each cassette **66a-d** is mounted to the device module **68a-d** in a vertical orientation. In other embodiments, each cassette **66a-d** may be mounted to the device module **68a-d** in other mounting orientations. Each cassette **66a-d** is configured to interface with and support a proximal portion of an EMD (not shown). In addition, each cassette **66a-d** may include elements to provide one or more degrees of freedom in addition to the linear motion provided by the actuation of the corresponding stage **62a-d** to move linearly along the linear member **60**. For example, the cassette **66a-d** may include elements that may be used to rotate the EMD when the cassette is coupled to the device module **68a-d**. Each device module **68a-d** includes at least one coupler to provide a drive interface to the mechanisms in each cassette **66a-d** to provide the additional degree of freedom. Each cassette **66a-d** also includes a channel in which a device support **79a-d** is positioned, and each device support **79a-d** is used to prevent an EMD from buckling. A support arm **77a**, **77b**, and **77c** is attached to each device module **32a**, **32b**, and **32c**, respectively, to provide a fixed point for support of a proximal end of the device supports **79b**, **79c**, and **79d**, respectively. The robotic drive **24** may also include a device support connection **72** connected to a device support **79**, a distal support arm **70** and a support arm **77o**. Support arm **77o** is used to provide a fixed point for support of the proximal end of the distal-most support arm **79a** housed in the distal most device module **32a**. In addition, an introducer interface support (redirector) **74** may be connected to the device support connection **72** and an EMD (e.g., an introducer sheath). The configuration of robotic drive **24** has the benefit of reducing volume and weight of the drive robotic drive **24** by using actuators on a single linear member.

(46) To prevent contaminating the patient with pathogens, healthcare staff use aseptic technique in a room housing the bedside unit **20** and the patient **12** or subject (shown in FIG. 1). A room housing the bedside unit **20** and patient **12** may be, for example, a cath lab or an angio suite. Aseptic technique consists of using sterile barriers, sterile equipment, proper patient preparation, environmental controls and contact guidelines. Accordingly, all EMDs and interventional accessories are sterilized and can only be in contact with either sterile barriers or sterile equipment. In an embodiment, a sterile drape (not shown) is placed over the non-sterile robotic drive **24**. Each cassette **66a-d** is sterilized and acts as a sterile interface between the draped robotic drive **24** and at least one EMD. Each cassette **66a-d** can be designed to be sterile for single use or to be re-sterilized in whole or part so that the cassette **66a-d** or its components can be used in multiple procedures.

(47) As previously discussed, embodiments of a control station **26** can include a variety of different input systems for controlling the bedside unit **20**. Input systems can include a variety of different input controls (for example, buttons, scroll wheels, joysticks, etc.) that can be manipulated by a user to control the robotic drive **24**. These input controls can be arranged in different layouts or patterns on the input system so that a user can easily reach each of them without taking their hands off of the controls. This may be useful, for example, so that the user can simultaneously and independently control the movement of multiple different EMDs or device modules **32**. Additionally, embodiments of an input system can be configured to operate in a variety of different control modes. In each control mode, different functions can be assigned to each of the input controls based on, amongst other factors, the procedure being performed, which device or devices are being controlled, user preferences, or any other factors. The input system can be configured to switch between different control modes to reassign functions to at least one of the input controls in response to the user or the control computing system **34**.

(48) In some embodiments, the body of an input system can be configured to be held in the hands of a user as the user manipulates controls coupled thereto. The term handheld refers to the capability of being held by at least one hand of a user while the user manipulates one or more controls coupled thereto with the at least one hand. The term handheld includes but is not limited to having dimension and weight to facilitate carrying by a user and manipulation of one or more controls coupled thereto while being carried, whether or not removably mountable to a fixed support. The term handheld includes but is not limited to having a structure of dimension and/or weight which does not facilitate carrying by a user.

(49) An input system may be fixed to or integrated with a surface of the control station **26**, or an input system may be configured to selectively operate in one of two or more of these arrangements. FIG. 4 illustrates a top-down view of a handheld input system **100** for controlling a catheter-based procedure system to perform a catheter-based medical

procedure in accordance with an embodiment. The handheld input system **100** includes a body **104** with a left extension **108** and a right extension **110** extending from opposite lateral sides of the body **104**. The left and right extensions **108**, **110** are configured to be held by the user's left and right hands, respectively, and function as handles or handle portions to enable secure gripping of the handheld input system **100** by the user. A top surface **114** and a front surface **116** of the handheld input system **100** include various input controls **120** that are arranged so that they can be manipulated, actuated, or otherwise interacted with by at least one of the digits of the left and right hands of the user while the user holds the body **104**.

(50) The illustrated handheld input system **100** includes two groups of input controls **120**—a left input array **122** and a right input array **124**—arranged on the top surface **114** proximate the left and right sides of the body **104**. When the handheld input system **100** is held by a user, the left and right input arrays **122**, **124** are respectively positioned within the range of motion of the left and right thumbs of a user. Similarly, input controls **120** positioned on the front surface of the handheld input system **100** are grouped in a left shoulder group **128** and a right shoulder group **130** that are respectively within the range of motion the left and right index fingers of the user. Thus, the left input array **122** is configured so that it can be manipulated by a thumb of a left hand of the user, and the shoulder input group **128** is configured so that it can be manipulated by the index fingers and/or the middle fingers of the left hand of the user simultaneously with manipulation of input array **122** by the thumb of the left hand. Similarly, the right input array **124** is configured so that it can be manipulated by a thumb of a right hand of the user, and the shoulder input group **130** is configured so that it can be manipulated by the index fingers and/or the middle fingers of the right hand of the user simultaneously with manipulation of input array **124** by the thumb of the right hand.

(51) It should be noted that the any input system described herein may be of dimension and weight to be carried by a user or integrated with the control station **26**. An input system configured to be carried by a user may also be configured for removable mounting to the control station **26** or other fixed support.

(52) In some embodiments, handheld input systems can include additional input controls that can be grouped with, or separate from, other input controls **120**. For example, the top surface **114** of the illustrated handheld input system **100** further includes multiple input controls **120** that are positioned centrally with respect to the body **104**, and a directional input control **134** positioned proximate the right input array **124** so that it can be manipulated by the right thumb of the user. The directional input controls **134** may be a knob, a joystick, a directional pad, a touch pad, or any other input control that can be manipulated by a user to command the robotic drive **24** to move an EMD and/or a device module **32** in multiple different directions. In other embodiments, however, another type of input control may be used. Additionally, the handheld input system **100** includes a touchpad input control **136** positioned between the left and right input arrays **122**, **124** proximate the front side of the top surface **114**.

(53) A variety of different input control types and configurations may be used for each of the input controls on embodiments of handheld input system. Referring to FIG. 5, the handheld input system **100** can include at least one binary input control, such as binary buttons **140**, which may be arranged in various locations on the top surface **114** and the front surface **116**. The left and right input arrays, for example, are configured as left and right button arrays **142**, **144** that each include multiple binary buttons **140**. Additionally, the left and right shoulder groups **128**, **130** each include a binary shoulder button **141** proximate a top side of the front surface **116**, and three binary buttons **140** are positioned on the top surface **114** between the two button arrays **142**, **144**. When manipulated by the user, each of the binary buttons **140** can send a control signal indicating that that binary button **140** has been pressed. For example, at least one binary button **140** can be configured to send a control signal while the binary button **140** is held in a pressed state by the user, or whenever the binary button **140** is switched between the pressed and unpressed state.

(54) Some handheld input systems can include at least one analog input control that sends a control signal which may vary between a minimum and a maximum value. In the illustrated embodiment, for example, the left shoulder group **128** and the right shoulder group **130** each include an analog trigger **148** arranged on the front surface **116** of the handheld input system **100** below the corresponding one of the shoulder buttons **141**. The analog triggers **148** are configured to be pulled by at least one of the user's fingers through a range of positions between an unpulled (or unpressed) position and a fully-pulled (or fully-pressed) position, and each analog trigger **148** can include a biasing element (not shown) that biases the analog trigger **148** into the unpulled position once it is released by the user. Additionally, each analog trigger **148** is configured to send a control signal that varies based on the position of the analog trigger **148**. For example, at least one analog trigger **148** can be configured to send a control signal that has a first value when the analog trigger is in the unpulled position, a second value when the analog trigger is in the pulled position, or a third value that can be interpolated between the first and second values based on when the analog trigger **148** is relative to the unpulled and fully-pulled positions.

(55) In some embodiments, an analog input control can be configured as a tilting joystick **150** that can be manipulated by the user's right thumb. FIG. 6 illustrates a detailed view of the joystick **150** used with the handheld input system **100** of FIG. 5. In addition to providing a control signal that varies based on how far the joystick **150** is tilted away from its vertical resting position, the control signal may also vary based on the direction that the joystick **150** is tilted. For example, the joystick **150** may be configured to send a control signal that indicates how far and in what direction a user has moved the joystick **150**. In some embodiments, a joystick (or other directional analog input) may be

configured to send a control signal that only indicates the direction that it is tilted in or how far it has been tilted. A joystick may also be configured to only send a control signal when the joystick is tilted in one or more specific directions or ranges of directions. Additionally or alternatively, the control computing system **34** may be configured to only process or act based on certain aspects of a control signal, for example the direction or tilt angle of a joystick. (56) Some handheld input systems can include at least one input control configured as a scrolling or rotating input control. For example, the handheld input system **100** of FIG. **7** includes a scroll wheel **154** positioned on the top surface **114** and configured to be manipulated by the right thumb of the user. Such manipulation by the right thumb may occur simultaneously with manipulation of right shoulder button **141** and/or right analog trigger **148** with one or more fingers of the user's right hand. Manipulation by the right thumb and one or more fingers of the user's right hand may further occur simultaneously with manipulation of left shoulder button **141** and/or left analog trigger **148** with one or more fingers of the user's left hand.

(57) FIG. **8** provides a detailed view of the scroll wheel **154** of FIG. **7**. The scroll wheel **154** is partially enclosed within the handheld input system **100** so that a portion of the scroll wheel **154** projects outward from the top surface **114** and is accessible to the user, while a different portion of the scroll wheel is positioned within the body **104**. The axis of rotation of the scroll wheel **154** is generally parallel to the top surface **114** so that, when it is rotated by a user, part of the exposed section of the scroll wheel **154** moves into the body **104** while part of the hidden section of the scroll wheel **154** moves out of the body **104** into view of the user. As the user rotates the scroll wheel **154** (or another scrolling input control), it provides a control signal which relays how far and in which direction the scroll wheel **154** is rotated to the control computing system **34**.

(58) In some embodiments, a scroll wheel **154** can include a plurality of detents configured to provide discrete rotational positions for the scroll wheel **154**. As the scroll wheel **154** is rotated between these positions, the detents may provide the user with tactile feedback. For example, the detents may provide initial resistance against rotation of the scroll wheel **154**, but can then push the scroll wheel **154** once it is rotated far enough, thereby causing it to jump or "pop" into the next rotational position. The scroll wheel **154** can be configured to send a control signal each time it is moved between the rotational positions defined by the detents, providing a first control signal when the scroll wheel **154** is rotated in a first direction or a second control signal when it is rotated in a second direction opposite the first.

(59) Some embodiments of a handheld input system can include a scrolling input control in a different configuration. For example, FIG. **9** illustrates the handheld input system **100** with a scrolling input control configured as a jog wheel **156**, which is illustrated in detail in FIG. **10**. The jog wheel **156** is generally flat and has an axis of rotation that perpendicular to the top surface **114** so that, when rotated by the user, the jog wheel **156** spins without moving into or out of the body **104**. The illustrated jog wheel **156** includes a disk portion **158** that rotates around a hub **160**. In some embodiments, the hub **160** can be configured as an input control that may be pressed by the user. FIG. **11** illustrates another scrolling input control configured as a dial **162**. As with the jog wheel **156** of FIGS. **9** and **10**, the dial **162** has an axis of rotation that is perpendicular to the top surface **114** of the handheld input system **100**. The dial **162** may project away from the top surface to a raised face **163**, which can include a recess **164** configured to be engaged by a user's finger to rotate the dial **162**. As illustrated in FIG. **12**, a scrolling input control can additionally be configured as a knob **166** that can be manipulated by at least one digit of a user's hand.

(60) Further still, a handheld input system can include more than one scrolling input control. FIG. **13** illustrates a handheld input system **100** that includes two scroll wheels **154**. The first scroll wheel **154** is arranged on the top surface **114** and is configured to be manipulated by a right thumb of a user, and the second scroll wheel **154** is positioned on the front surface **116** and is configured to be manipulated simultaneously, if desired, by a different finger, on the user's right hand, such as the index finger or the middle finger. This may be useful, for example, in order to control two degrees of freedom of an EMD with scrolling input controls (as described in reference to FIGS. **16** and **17**), and/or one degree of freedom of each of two different EMDs.

(61) FIG. **14** illustrates a handheld input system **100** that includes two jog wheels **176**, **178** that may be manipulated by a user's left and right thumbs, respectively and simultaneously, if desired. The FIG. **14** arrangement may be useful, for example, in order to control two degrees of freedom of an EMD, and/or one degree of freedom of each of two different EMDs.

(62) Some embodiments of a handheld input system can include a touch input control (such as the touchpad **136**), which may be configured as at least one of a binary input control, an analog input control, and a scrolling input control. For example, a capacitive touch pad can be configured as a directional swipe pad that may be used as an alternative to a physical joystick. Embodiments of a jog wheel can be configured as a mechanical component that can be physically rotated by the user, or as a touch input control, such as a capacitive touch pad, can be used to simulate a mechanical jog wheel. Similarly, a touch input control can be used to simulate a scroll wheel.

(63) Embodiments of a handheld input system can be configured to control a catheter-based procedure system in a variety of different modes that use various combinations of input controls as motion controls or selection controls. When used as a motion control, an input control can be configured to control axial and/or rotational movement of at least one EMD. Further, input controls that are used as motion controls may be configured to control movement of an

EMD in a position control mode or a speed control mode.

(64) Any of the above-described input controls, when used in a position control mode, can instruct the robotic drive **24** to actuate an EMD by a prescribed increment when the input control is activated. When controlling axial movement in a position control mode, an input control can command at least one device module **32** to move a discrete distance in the distal or proximal direction. When controlling the rotational movement of an EMD in a position control mode, an input control can command a device module **32** to rotate the EMD a discrete angle in the clockwise or counterclockwise direction. When a position motion command is issued in a closed-loop system, the control computing system **34** can compare the commanded increment or position with a measured increment or position. If the commanded value is different than the measured value, the control computing system **34** can close the loop by providing an additional motion command to correct the difference.

(65) When used in a speed control mode, any of the above-described input controls can instruct the robotic drive **24** to continuously actuate an EMD at a prescribed rate while the input control is activated by the user. When controlling axial movement, an input control in speed control mode can command at least one device module **32** to move in the distal or proximal direction continuously at the prescribed rate until the user deactivates the input control (or a limit of the robotic drive **24** is reached). Similarly, an input control in speed control mode can command at least one device module **32** to continuously rotate at the prescribed rate as long as the input control is activated. When a speed motion command is issued in a closed-loop system, the control computing system **34** compares the commanded speed with the measured speed and, if a difference is detected, adjust the movement speed to close the loop.

(66) With continued reference to FIG. **14**, embodiments of the handheld input system **100** can be configured to control the robotic drive **24** of a catheter-based procedure system using multiple control modes. The handheld input system **100** includes two analog input controls configured as a left analog trigger **170** and a right analog trigger **172**, and two scrolling input controls configured as a left jog wheel **176** and a right jog wheel **178**. Additionally, the handheld input system **100** includes multiple binary buttons **140** that are grouped into a left or right button array **142**, **144**, configured as a left shoulder button **182** or a right shoulder button **184**, or arranged proximate the center of the top surface **114**. Each of the binary buttons **140**, analog triggers **170**, **172**, and jog wheels **176**, **178** can be configured to operate in a speed control mode and a position control mode.

(67) When used in a position control mode, the binary buttons **140** may be configured to instruct the robotic drive **24** to actuate a selected EMD or device module **32** by the prescribed increment each time the user activates binary button **140**. For example, the left shoulder button **182** can be configured to command the robotic drive **24** to actuate (axially or rotationally) an EMD in a first direction by a prescribed increment and the right shoulder button **184** can be configured to actuate the EMD in a second, opposite direction by the same prescribed increment. In speed control mode, the binary buttons **140** device can be configured to instruct the robotic drive **24** to begin actuating an EMD to move axially or rotate at a prescribed rate pressed by the user, and can continue to do so until the user releases the binary button **140** or a limit of the robotic drive **24** is reached. Similar to how they function in a position control mode, left and right shoulder buttons **182**, **184** can be configured to command a robotic drive **24** to actuate (axially or rotationally) an EMD in a first direction and or a second direction opposite the first.

(68) The left and right analog triggers **170**, **172** can be configured to instruct the robotic drive **24** to move or rotate a selected EMD or device module **32** by a variable increment when used in a position control mode, or at a variable rate when a speed control mode. When one of the analog triggers **170**, **172** is fully pulled by the user, it may command the robotic drive **24** to actuate (axially or rotationally) a selected EMD by the full value of a prescribed increment. If one of the analog triggers **170**, **172** is partially pulled by the user, however, it may command the robotic drive **24** to actuate the EMD by a fraction of the prescribed increment corresponding to how far the trigger was pulled. When in a speed control mode, the analog triggers **170**, **172** may similarly be configured to instruct the robotic drive **24** to actuate (axially or rotationally) an EMD at a fraction of a full prescribed rate when the analog trigger **170**, **172** is partially pulled by a user. In some embodiments, the left analog trigger **170** can be configured to command the robotic drive **24** to move or rotate an EMD in a first direction, while the right analog trigger **172** can be configured to command the robotic drive **24** to move or rotate an EMD in a second direction opposite the first. Thus, when used together, the left and right analog triggers **170**, **172** can provide speed controls and/or position controls for moving or rotating an EMD. Additionally or alternatively, the left and right analog triggers **170**, **172** can be configured to command the robotic drive **24** to move or rotate an EMD in the same direction, but at different rates.

(69) As with binary and analog input controls, scrolling input controls can be configured in a position or speed control mode. When configured in a position control mode, the left and right jog wheels **176**, **178** can be configured to instruct the robotic drive **24** to move or rotate an EMD by a discrete increment each time the user rotates the jog wheels **176**, **178** a nominal angular distance. For a jog wheel **176**, **178** that includes detents, this nominal angular distance may correspond to the angular distance between the discrete rotational positions created by the detents. Thus, the robotic drive **24** to move or rotate an EMD by the prescribed increment each time the user rotates the left and right jog wheels **176**, **178** between rotational positions. If a jog wheel **176**, **178** is moved between multiple rotational positions, the total commanded rotational or angular motion may be determined by multiplying the prescribed increment by the number of times the user moved the jog wheel **176**, **178** between different rotational positions. The

direction of the commanded EMD motion may depend on the direction that the left or right jog wheel **176, 178** is rotated. Rotating a jog wheel **176, 178** in a clockwise direction may instruct the robotic drive **24** to move or rotate the EMD in a first direction, and rotating the jog wheel **176, 178** in a counterclockwise direction may instruct the robotic drive **24** to move or rotate the EMD in a second direction opposite the first direction.

(70) When configured in a speed control mode, a scrolling input control may function similarly to a throttle, where rotation of the scrolling device in one direction increases a commanded rate of EMD motion, and rotation in the opposite direction decreases the commanded rate of EMD motion. In the illustrated embodiment, the left and right jog wheel **176, 178** can have a neutral position in which they do not provide an instruction for the robotic drive **24** to move or rotate and EMD. Each time one of the jog wheels **176, 178** is rotated a nominal angular distance it can instruct the robotic drive **24** to either increase or decrease a rate of axial or rotational movement based on the direction that the user rotates the jog wheel **176, 178**. The robotic drive **24** can continue to move or rotate the EMD at the commanded rate until the user adjusts the rate by moving the jog wheel **176, 178** to a different rotational position. To stop the robotic drive **24** from moving or rotating the EMD, the user can return the corresponding jog wheel **176, 178** to the neutral position.

(71) Although the use of binary input controls, analog input controls, and scrolling input controls in position and speed control modes has been described in reference to the binary buttons **140**, analog triggers **170, 172**, and jog wheels **176, 178** in FIG. **14**, it should be appreciated that some embodiments of a handheld input system can include at least one input control that is different than those of the illustrated embodiments. Additionally or alternatively, a handheld control module can include input controls that are configured in a different arrangement than those of the illustrated embodiments.

(72) In some operating modes, the relationship between the binary, analog, and scrolling input controls and the increment of commanded axial movement or rotation of an EMD may be fixed. When operating in such a mode, the value of the prescribed increments and the prescribed rates for position controls and speed controls, respectively, may be fixed for an input control. In other modes of operation, however, the relationship between the binary, analog, and scrolling input controls and the amount of commanded axial movement or rotation of an EMD may be configurable by the user or by the control computing system **34**. For example, the handheld input system **100** of FIG. **14** can include at least one binary button **140** configured as a scaling input that can adjust (for example, increase or decrease) the prescribed increment or the prescribed rate associated with at least one input control. A handheld input system **100** with one binary button **140** configured as a scaling input can be configured to cycle between multiple preset options for the prescribed increments and rates for at least one input control.

(73) A handheld input system **100** can include multiple binary buttons **140** that are configured as scaling inputs for adjusting the prescribed increment and rate for one input control or group of input controls. The left button array **142** or the right button array **144** can include two binary buttons **140** configured as scaling inputs—one binary button **140** being configured to increase the prescribed increment and rate associated with the associated input control (s), the other binary button **140** being configured to decrease the prescribed increment and rate. For example, when configured to adjust movement instructions issued by the jog wheels **176, 178** (or any other scrolling input control) in a position control mode, a binary button **140** scaling input may adjust the amount of movement that is commanded each time the jog wheels **176, 178** are moved between the rotational positions created by the detents.

(74) The scaling inputs can be configured so that, each time one is activated by the user, the prescribed rate and prescribed increments of the associated input control (s) may be increased or decrease by a predetermined value. Additionally or alternatively, the prescribed increment or rate of an input control may be increased or decreased based on a scaling factor each time a corresponding scaling inputs is pressed. For example, a handheld input system **100** can be configured so that the prescribe rate associated with an analog triggers **170, 172** in a speed control mode is multiplied by a scaling factor (greater than one to increase the prescribed rate or less than one to decrease the prescribed rate) when one of the scaling inputs is pressed. In some embodiments, a handheld input system **100** can include at least binary button **140** that is associated with a specific scaling factor. While that binary button **140** is held by the user, motion commands issued using the handheld input system **100** may be multiplied by the associated scaling factor.

(75) In some embodiments, a handheld input system can be configured with selection verification features that require a user to confirm that the desired EMD and/or device module **32** has been selected before any motion commands are sent to the robotic drive **24**. In the illustrated embodiment, at least one of the binary buttons **140** on the handheld input system **100** can be configured as a selection control that can be used to select which of the device modules **32** and/or which of the EMDs will be controlled by the motion controls. For example, the left button array **142** or the right button array **144** can be configured as a selection array in which each of the binary buttons **140** corresponds to a single EMD or device module **32**, or to a group of multiple EMDs and/or device modules **32**. Each of the selection controls in a selection array may be arranged in a logical order coordinating with the EMD's orientation in use, which may mimic their orientation in manual intervention. For example, the selection controls may be arranged based on their size, where the device with the largest diameter (for example, a guide catheter) corresponds to the leftmost selection control, the device with the smallest diameter (for example, a guide wire) corresponds to the rightmost

selection control, and any selection controls corresponding to other EMDs are positioned in between.

(76) While some handheld input systems **100** can be configured to control an EMD or device module **32** after the corresponding selection control is momentarily pressed, other embodiments made be configured in a continuous activation mode that requires a user to hold a selection control in order for motion commands to be sent to the corresponding device module **32**. If the user tries to use a motion control to move an EMD or device module **32** without holding the selection control, the robotic drive **24** would not move any devices. Additionally, the robotic drive **24** may stop moving a device if the user releases the corresponding selection control while the device is moving because a corresponding instruction would no longer be sent to the robotic drive **24**. This may be useful, for example, in order to help prevent incidental movement of an EMD or device module **32**. A single selection control may be associated with one or more input controls such that it controls all instructions sent as a result of manipulation of the one or more input controls.

(77) In order to select multiple EMDs or device modules **32** simultaneously, a user can hold multiple selection controls simultaneously. Some selection controls can be configured to be assigned to specific device combinations of EMDs and/or device modules **32**. Additionally or alternatively, a handheld input system **100** could include a binary button **140** configured as a grouping button that, when held, causes the control computing system **34** or the handheld input system **100** to store or record any selection controls that are pressed while the grouping button is held even after those selection controls are released. While the user continues to hold the grouping button, the handheld input system **100** can provide instructions to move or rotate any of the devices that are associated with the stored selection controls. When the grouping button is released by the user, the stored selection controls may be cleared.

(78) In some embodiments, a directional input control, such as a joystick or directional pad, can be configured as a selection input control. For example, the positions around the joystick (e.g., up, down, left, and right) may correspond to different EMDs, device modules **32**, or groups of devices. To select a device or group of devices, the user can manipulate the joystick to point to the desired device(s). Pointing the joystick between two positions would select the EMDs and/or device modules **32** corresponding to both positions. In another embodiment, an EMD or device module **32** can be selected by invoking an on-screen selection menu, and choosing one or more of the presented options. The input controls used for navigating the menu may be also be used as motion controls, while some embodiments can include separate input controls that are only used for menu navigation.

(79) Using the button arrays **142**, **144** and shoulder buttons **182**, **184**, the analog triggers **170**, **172** and the jog wheels **176**, **178** as motion controls, scaling inputs, and selection controls in various combinations of the above configurations, the handheld input system **100** of FIG. **14** can operate in various different control modes. In some embodiments, the handheld input system **100** may switch between control modes in response to information from the computing control computing system **34**. For example, the handheld input system **100** can be configured to operate in different control modes during different parts of a procedure. Additionally or alternatively, the handheld input system **100** can switch control modes based on which EMDs or device modules **32** are selected, or in response to a user manually switching control modes.

(80) In some control modes, a user can control axial movement of a selected EMD by manipulating the analog triggers **170**, **172** and/or the shoulder buttons **182**, **184** with their index fingers or middle fingers. In this control mode, the left button array **142** may be configured as a selection array, and the user may select an EMD or device module **32** to control. The handheld input system may be configured so that the right analog trigger **172** and the right shoulder button **184** (the right shoulder group **130**) can instruct the robotic drive **24** to move the EMD in the distal direction, and the left analog trigger **170** and the left shoulder button **182** (the left shoulder group **128**) can instruct the robotic drive **24** to move the EMD in the proximal direction. The left and right analog triggers **170**, **172** can be configured in a speed control mode, and can therefore command variable speed axial motion based on how far the analog triggers **170**, **172** are pulled by a user. Additionally, two binary buttons **140** in the right button array **144** may be configured as scaling inputs for adjusting the rate of axial motion commanded by the analog triggers **170**, **172**. One of the scaling inputs can be configured to increase the axial rate of motion (for example, by doubling the speed), while the other scaling input can be configured to reduce the axial rate of motion (for example, by halving the speed). An increased speed may be where the vasculature is not tortuous and has side branches that are easily avoided, and an even higher speed may be used for moving EMDs within other EMDs.

(81) In some embodiments, and, for example, in place of right analog trigger **172**, an analog rocker could be sprung to the center of its range, where manipulation in one direction commands axial motion in the distal direction, while manipulation of the rocker in the opposite direction commands axial motion in the proximal direction. Additionally, a binary button, for example a shoulder button, may be configured to reverse the direction of motion commanded by an analog trigger.

(82) In addition to speed controls, the left shoulder button **182** and the right shoulder button **184** can be configured as position controls for moving the selected EMD in the proximal and axial directions, respectively. Additionally or alternatively, two binary buttons **140** in the right button array **144** may be configured as position controls for commanding axial movement of and EMD. In some control modes, the speed control and the position controls may be simultaneously available to the user, and in some control modes only one type of control will be available at a

given time. Rotation of the EMD may be linked to at least one of the left and right jog wheels **176, 178**, which may be manipulated by the corresponding thumbs of the user in order to instruct the robotic drive **24** to rotate the EMD. The jog wheels **176, 178** can be configured in a position control mode, and in some embodiments, the scaling inputs may be pressed by the user to adjust the relationship between rotation of the jog wheels **176, 178** and the rotation of the EMD.

(83) In some control modes, a user may be able to simultaneously and independently control axial motion and rotational motion of two different EMDs using the thumbs and fingers of both hands as described above. For example, the handheld input system **100** can be configured so that the input controls on the left half of the handheld input system **100** control movement of one EMD, while input controls on the right half of the handheld input system **100** control movement of a second EMD. In some control modes, the left button array **142** and the right button arrays **144** may be configured as selection arrays for selecting which EMD or device module **32** will be controlled by the left and right sides of the handheld input system, respectively. In another control mode, the input controls on one side of the handheld input system **100** can be permanently assigned to an EMD or device module **32**.

(84) Input controls on the left and right shoulder groups **128, 130** can be configured to independently control axial movement of two different EMDs. For example, the analog triggers **170, 172** can be configured to instruct the robotic drive **24** to move the respective EMDs in a distal direction, while the shoulder buttons **182, 184** can instruct the robotic drive **24** to move the respective EMDs in a proximal direction. Rotation of the selected EMDs may be independently controlled using the left and right jog wheel **176, 178**. In some embodiments, a handheld input system may only include one jog wheel, which may be configured to control rotation of both of the selected EMDs, or it can be configured to only control rotation of one of the EMDs.

(85) According to some embodiments, the robotic system is configured so that the input controls instructing motion of the wire-based EMD are mapped to the device module holding the wire-based EMD. The identity of the device module holding the wire-based EMD may be detected with sensors in response to loading the EMD into the device module. Detection may employ contact or non-contact sensors, such as mechanical, electrical or visual sensors, or by a user input prompted by the system.

(86) The controls on the right side of the input system **100** instruct motion of the device module which is holding the wire-based EMD. For example, jog wheel **178** can be rotated by the user's right thumb to rotate the wire-based EMD. Trigger **172** advances the wire-based EMD at a speed which corresponds with the degree to which the trigger is pulled by the user's right index finger. Shoulder button **184** can be pressed by the user's index finger to retract the wire-based EMD. Jog wheel **178** and either of trigger **172** and button **184** may be manipulated simultaneously.

(87) The controls coupled to the left side of the input system **100** control a selected catheter. Button array **142** includes buttons **140E, 140F** and **140G**, and pressing one of the buttons (typically with the left thumb) selects a device module **1, 2** or **3** respectively such that manipulation of the controls on the left side of the input system **100** results in the sending of instructions to the selected device module. The input system **100** may allow multiple catheters to be selected by pressing the selection buttons **140E, 140F** and **140G** in series. Pressing button **140H** may deselect any selected device modules. The left side of the input system **100** further includes controls **176, 170** and **182** to instruct the selected device module(s) to control the motion of the selected catheter, in the same way as the corresponding controls described above on the right side (e.g., using a left index finger and left thumb, simultaneously or otherwise).

(88) In another embodiment, the controls for the wire-containing device module are not separated from controls of the catheter-containing device modules. For example, scroll wheel **154** is positioned on the upper right shoulder of input system **100** of FIG. **13** to be manipulated by the index finger of the right hand for linear motion. Four selection buttons **140A, 140B, 140C, 140D** are arranged for device module selection by the right thumb, with each button corresponding to one of catheter **1, 2, 3** and the wire-based EMD. In one embodiment, tapping and releasing a button **140A, 140B, 140C** or **140D** selects the associated device module. In another embodiment, the user is required to hold depressed the button or buttons associated with the selected device modules to maintain selection thereof.

(89) Scroll wheel **154** on top surface **114** of FIG. **13** may be manipulated by the left thumb in order to control rotation of the device selected by the right thumb. The user can move their left hand off the scroll wheel **154** and select a device module with buttons **140E, 140F**, and/or **140G**, respectively assigned to Catheter **1, Catheter 2** and Catheter **3**. When selecting a device module with the left thumb, the left index finger can manipulate buttons **141** and **148** to control linear motion of the left hand-mapped device, while the right thumb simultaneously selects another device or devices using buttons **140A, 140B, 140C, 140D** and the right finger manipulates scroll wheel **154** to control linear motion of that device.

(90) The same functionality and control scheme as described above for FIG. **13** can be applied to similarly-arranged controls in different housings. In another example, FIG. **13B** shows the same functional controls as in FIG. **13**, coupled to a handheld body designed to sit on or be integrated with a table/desk/console-type support. In one embodiment of operation, the linear motion control for the right hand-selected device(s) (i.e., selected by tapping and releasing one or more of buttons **140A, 140B, 140C** and/or **140D**, or by continuously depressing and holding the button(s) associated with the selected device modules) is right side scroll wheel **154** and the rotation control for the

right hand-selected device is now the knob **178** located near the center of the support and manipulable by a finger or fingers on the left hand. To perform tasks described above which require two-independently controlled linear degrees of freedom of two EMDs, the user moves their hand from the knob **178** to the controls on the left, selects a device module using the left thumb and buttons **140E**, **140F** or **140G** (either by tapping or by holding down associated button(s) **140E**, **140F** and/or **140G**), and manipulates the left side scroll wheel **154** to instruct linear motion of a second EMD, while the right hand continues to operate as described above.

(91) The following table describes operational modes of the FIG. **14** input system and the functions of various input controls in each operational mode according to some embodiments. Any module selection described in the table may require continuous activation of the corresponding selection button(s). One or more of the operational modes may be implemented by other input systems described herein.

(92) TABLE-US-00001 Operational Controls Buttons Control Controls Buttons Control Mode 170/182 140E-140H 176 172/184 140A-140D 178 Single EMD not used Module 1-4 not used Linear motion not used Rotation selection of selected of selected module module Two EMDs Linear motion Module 1-3 Selected Linear motion no selection Rotation of catheter selection catheter of wire required - of wire rotation wire selected 1-device mode not used not used Rotation Linear motion Module 1-4 not used of selected of selected selection 2-device mode Linear motion Module 1-3 not used Linear motion Module 1-4 not used of secondary secondary of primary primary selection selection

(93) Below, an example endovascular treatment of acute ischemic stroke using combined technique (i.e., using a stent retriever with aspiration) is described using the handheld input system **100** shown in FIG. **14**. The introducer sheath is manually inserted into the femoral artery, giving a passageway for all other interventional devices (EMDs) to access the vasculature. The guide catheter, diagnostic catheter and guidewire are coaxial and inserted into the introducer sheath. To reduce risk of vessel injury, the EMDs are not inserted beyond the distal end of the introducer sheath. At this point, these EMDs are loaded into the robotic system, with the guide catheter in module **1**, the diagnostic catheter in module **2** and the guidewire in module **3**.

(94) The interventionalist can observe the position of the EMDs in the body via fluoroscopic imaging. Using the input module, the EMDs are navigated up the femoral artery through the descending aorta to the aortic arch with the guidewire leading, as follows. Using the input system **100**, the guidewire is selected as the active device and the analog trigger **172**, a linear position control manipulated by the user's index finger, is used to instruct the system to advance the guidewire at a speed corresponding to the amount that the trigger is pulled. The advancing control could also be a scroll wheel which operates as a positional control, as described above and shown as element **154** in FIG. **13**. Jog wheel **178** is manipulated rotationally by the user's right thumb to direct the angled tip of the wire along the correct path, and not into any side branching vessels. With the guidewire setting the path, the diagnostic and guide catheters are selected by activating devices **1** and **2**, using the left thumb to depress **140E** and **140F** in series. Control **170** may then be depressed to advance the diagnostic and guide catheters along the guidewire to the base of the arch.

(95) In the aortic arch, the guidewire and diagnostic catheter are manipulated sequentially or at the same time to direct the tip of the EMDs into the desired internal carotid artery branching off of the aortic arch. This might require the interventionalist to rotate and push the guidewire at the same time to get the tip in the right location, which can be accomplished by simultaneously manipulating jog wheel **178** with the right thumb along with one of either **184** or **172** linear controls with the right index finger. With the vessel selected, the guidewire can be advanced linearly, using rotation control as needed to stay on the desired path.

(96) Once the guidewire has moved several inches up into the internal carotid artery, the diagnostic catheter and guide catheter can follow. This may be done by selecting both the diagnostic catheter and guide catheter as the active devices by pressing device selection control buttons **140E** and **140F** in series, and manipulating, with the left index finger, analog trigger **170** to instruct drive modules **1** and **2** to linearly move their respective EMDs simultaneously. With the guide catheter in the internal carotid artery, the diagnostic catheter and guidewire are removed by using index fingers to manipulate control **184** to retract the wire, pressing **140F** to select the diagnostic catheter and then pressing control **182** with the left index finger to retract the selected catheter, until both are safely within the guide catheter. At that point, the bedside operator can remove the diagnostic catheter and guidewire from the system manually. An aspiration catheter, microcatheter and microwire are then inserted into the guide catheter and then loaded in device modules **2**, **3** and **4**, respectively.

(97) Control is assumed again by the interventionalist, who may simultaneously select to drive the aspiration catheter and microcatheter in modules **2** and **3** respectively by pressing **140F** and **140G** in series with the left thumb and holding button **140H** to activate high speed travel mode for the analog trigger **170**. The microwire is advanced by pulling trigger **172** with the right index finger. High speed travel mode may be used if devices are traveling within another catheter. The user releases button **140H** to return to normal speed travel mode before the devices reach the tip of the catheter in which they are traveling.

(98) In some instances, where tortuosity is encountered, the interventionalist may simultaneously select multiple catheters using some combination of **140E**, **140F** and **140G** pressed in series and commanding them to retract using control **182**, which has the effect of straightening bends in the vasculature for easier passage. In this case, the

interventionalist may desire simultaneous and independent control of the microwire, such that the position of the microwire can actively be compensated as the catheters are retracted to ensure that the microwire's placement is not lost in a specific vascular branch or that the microwire does not advance to an unsafe location. The microwire linear position is controlled by controls **172** and **184**, which can be manipulated by the right index finger simultaneously and independent from controls **170** and **182**, which are manipulated by the left index finger to control linear motion of the selected catheters.

(99) Once the target lesion is reached, the microwire and microcatheter are advanced through the clot. The microwire is retracted robotically inside the microcatheter, and then can be removed from the system. A stent retriever inserted into the microcatheter and its proximal end is loaded into device module **4**. The system detects the presence of the stent retriever device in device module **4** using techniques as described above and the system maps control of device module **4** to the right side of the input module, to be manipulated using the controls **178**, **184** and **172**. The stent retriever is advanced using **172** to properly position the stent retriever relative to the clot. The interventionist deploys the stent retriever by selecting the microcatheter by pressing **140G** and slowly retracting it by manipulating button **182** while simultaneously slightly compensating with the linear wire controls **184** and **172** (which instructs device module **4** holding the stent retriever) to keep the stent retriever in place while the microcatheter is retracted into the aspiration catheter. The aspiration catheter is advanced to the face of the clot by pressing **140F** and advancing using trigger **170**. Tubing is connected to the hub of the aspiration catheter so that vacuum can be applied to the aspiration catheter. The stent retriever and the aspiration catheter are now retracted together, by selecting the aspiration catheter by pressing **140F** and simultaneously pressing controls **182** and **184** to retract the aspiration catheter and the stent retriever respectively. The aspiration catheter and the stent retriever are retracted robotically into the guide catheter, and then can be unloaded from the system and fully retracted manually.

(100) In some embodiments, a catheter-based procedure system **10** can include at least one other input system. For example, FIG. **15** illustrates a perspective view of a handheld scrolling input system **200** for controlling a catheter-based procedure system **10** to perform a catheter based medical procedure in accordance with an exemplary embodiment. The scrolling input system **200** includes a body **204** configured to be grasped by the hand of a user so that the user's thumb can rest on a side surface **208** and the user's index finger and middle finger can rest on a top surface **210**. The illustrated scrolling input system **200** has a generally flat bottom surface and is configured to rest on a surface while held by the user. In some embodiments, however, a scrolling input system can be configured to be held by a user without resting on any supporting surface.

(101) A scrolling input control, such as the scroll wheel **216**, can be positioned on the top surface **210** of the scrolling input system **200** proximate a front side of the body **204** so that it may be manipulated by the user's index finger or middle finger. Similar to other scroll wheels illustrated in FIGS. **7** and **13**, the scroll wheel **216** can be positioned partially within the body **204** in a vertical orientation. One part of the scroll wheel **216** can be positioned within the body **204** so that it is inaccessible, while a different part of the scroll wheel **216** is accessible to the user and projects away from the top surface **210**. Additionally, the scroll wheel **216** can be oriented so that its axis of rotation extends laterally from a left side of the body **204** to a right side, and is generally perpendicular to a median plane of the scrolling input system **200**. Using their index finger or middle finger, a user can rotate the scroll wheel in a forward direction so that a point on the exposed section of the scroll wheel **216** moves towards the front side of the body **204**, or in a backwards direction so that the point on the exposed section of the scroll wheel **216** moves towards the back side of the body **204**. In some embodiments, the scroll wheel **216** can include detents that create discrete rotational position for the scroll wheel **216**, and which may provide tactile feedback to the user.

(102) Similar to the scrolling input controls described in connection to the handheld input system **100** of FIGS. **4-14**, the scroll wheel **216** can be configured as a motion control that may be configured in a speed control mode or a position control mode, and can be used instruct the robotic drive **24** to move or rotate an EMD. For example, the scroll wheel **216** of the scrolling input system **200** can be configured to command axial movement of a selected EMD in a position control mode. In some embodiments, the user can instruct the robotic drive **24** to move the EMD in a distal direction by rotating the scroll wheel **216** towards the front side of the scrolling input system **200**, and rotating the scroll wheel **216** towards the back side of the scrolling input system **200** can instruct the robotic drive to move the EMD in a proximal direction. As with the previously described scrolling input controls, the scroll wheel **216** can be configured to instruct the robotic drive to move or rotate an EMD by a prescribed increment each time the scroll wheel **216** is rotated a nominal angular distance, which may correspond to the angular distance between the discrete rotational positions created by the detents. Further, the relationship between the rotation of the scroll wheel **216** and rotation of the EMD may be scaled up or down (increasing or decreasing the amount of commanded rotation of the EMD) by the control computing system **34**, or by the user.

(103) Additionally or alternatively, a scrolling input system can include binary buttons that may be configured to move or rotate an EMD in a position or speed control mode, or to alter to operation of other input controls. In FIG. **15**, for example, the scrolling input system **200** can include two binary buttons **220**, **222** positioned on the top surface **210** so that they can be manipulated by the user's index finger or middle finger. The illustrated binary buttons **220**, **222** are positioned proximate the scroll wheel **216**, with a first binary button being position in front of the scroll wheel

216 and a second binary button **222** being arranged behind the scroll wheel **216**. In other embodiments, however, a scrolling input system can include at least one binary input control in a different location.

(104) In some control modes, the binary buttons **220**, **222** can be configured as motion controls for controlling axial movement of the EMD in a speed control mode. In this configuration, the user may instruct the robotic drive **24** to continuously move the EMD in the distal direction at a prescribed rate by holding the first binary button **220**, while holding the second binary button **222** may instruct the robotic drive **24** to continuously move the EMD in the proximal direction. Alternatively, the binary buttons **220**, **222** may be configured as scaling inputs that can be pressed by a user to adjust the relationship between rotation of the EMD and rotation of the scroll wheel **216**. Some embodiments of the scrolling input system **200** can include a mode-switching binary button (not shown) configured to switch the scroll wheel **216** into a continuous motion mode. While the binary button is held by the user, rotating the scroll wheel **216** forwards or backwards can instruct the robotic drive **24** to continuously move the EMD in the respective axial direction without repeated rotation of the scroll wheel **216**. In some embodiments, repeated rotation of the scroll wheel **216** may instruct the robotic drive to increase the axial movement speed of the EMD. Once the mode-switching binary button is released, the robotic drive **24** can stop moving the EMD and the scroll wheel **216** may return to a position control mode.

(105) In some embodiments a scrolling input system can include at least one additional input control for controlling a second degree of freedom of an EMD. For example, FIG. **16** illustrates a perspective view of a scrolling input system **200** that includes a secondary scroll wheel **226** and two additional binary buttons **228**, **230** positioned on the side surface **208**. The secondary scroll wheel **226** can be positioned partially within body **204** and may have an axis of rotation that generally extends from the front side of the scrolling input system **200** to the back side. Thus, using their thumb, the user can rotate the secondary scroll wheel **226** in an upward direction (clockwise when viewed from behind) or in a downward direction (counterclockwise when viewed from behind).

(106) When used in some control modes, the secondary scroll wheel **226** can be configured in a position control mode for commanding rotational motion of the selected EMD. For example, rotating the secondary scroll wheel **226** in the clockwise direction may instruct the robotic drive **224** to rotate the EMD in a clockwise direction, while counterclockwise rotation of the secondary scroll wheel **226** may instruct the robotic drive **24** to rotate the EMD in the counterclockwise direction. Thus, using a scrolling input system with two scroll wheel **216**, **226**, a user can simultaneously and independently control two degrees of freedom of an EMD with scrolling input controls. Similar to the first scroll wheel **216**, the relationship between rotation of the secondary scroll wheel **226** and the commanded movement of the EMD may be configurable. In some modes, there may be a 1:1 relationship between angular motion of the EMD and rotation of the secondary scroll wheel **226**. In such an embodiment, the robotic drive **24** will rotate the EMD the same angular distance that the secondary scroll wheel **226** was rotated by the user. In other embodiments, however, the robotic drive **24** can be configured to rotate the EMD more or less than the secondary scroll wheel **226** is rotated by the user.

(107) The third binary button **228** and the fourth binary button **230**, which are respectively positioned above and below the secondary scroll wheel **226** in the illustrated embodiment, can be configured to be pressed by the thumb of the user, and may be configured to function similarly to the binary buttons **220**, **222** on the top surface **210**. For example, the third and fourth binary buttons **228**, **230** can be configured in a speed control mode, and can respectively command rotation of the EMD in the clockwise and counterclockwise directions when held by the user. Alternatively, the third and fourth binary buttons **228**, **230** can be configured as scaling inputs for adjusting the relationship between rotation of the secondary scroll wheel **226** and commanded rotation of the EMD.

(108) In some embodiments, a catheter-based procedure system **10** can include at least one additional input system configured to be used with a scrolling input system. As illustrated in FIG. **17**, for example, a button pad **240** can include multiple input controls, such as binary buttons **242**, that may be manipulated using a second hand of a user while their first hand operates the scrolling input system **200**. Each of the binary buttons can be configured as selection controls for selecting which of the EMDs and/or device modules **32** will be controlled by the scrolling input system **200**. In some embodiments, the binary buttons **242** can be configured as selection controls in a continuous activation selection mode. While the user presses at least one of the binary buttons **242**, axial and rotational motion of EMDs and/or device modules **32** that correspond to the pressed selection controls can be controlled using the input controls of the scrolling input system **200**. However, if the user tries to move an EMD or device module **32** module without holding the selection control, the robotic drive **24** would not move any devices. Additionally, the robotic drive **24** will stop moving a device if the user releases the corresponding selection control. In other embodiments, at least one selection control can be configured in a toggling selection mode in which a selection control does not need to be held by a user.

(109) Some button pads can include at least one additional binary, analog, and/or scrolling input control configured to control axial motion or rotation of at least one EMD. FIG. **24** illustrates an embodiment of a button pad **240** that includes a plurality of first binary motion controls **250** arranged above the binary buttons **242** (which are configured as selection controls), and a plurality of second binary motion controls **252** arranged below the binary buttons. Each of the first and second binary motion controls **250**, **252** can be configured to control axial or rotational motion of an

EMD corresponding to the binary button **242** that a particular binary motion control **250, 252** is positioned above or below. For example, the first binary motion controls **250** can be configured to command the robotic drive **24** to move or rotate a corresponding EMD in a first direction when held by the user, while the second binary motion controls **252** can command the robotic drive **24** to move or rotate the corresponding EMD in a second direction. Using one hand, the user can simultaneously hold at least one selection control **242** and at least one binary motion control **250, 252**. Using the other hand at the same time, the user can manipulate a scroll wheel **216, 226** and/or press one of the binary buttons **220, 222, 228, 230** on the scrolling input system **200**. Thus, a user can control motion of at least one EMD by holding a corresponding one binary motion control **250, 252** while simultaneously controlling movement of the selected EMD(s) using the scrolling input system **200**.

(110) In a similar embodiment illustrated in FIG. **25**, an embodiment of a button pad **240** that includes a plurality of scrolling motion controls **256** arranged above, and corresponding to, one the binary buttons **242** configured as selection controls. As with the binary motion control **250, 252**, a user can manipulate the scrolling motion controls **256** to control axial or rotational motion of an EMD while simultaneously controlling motion of another EMD with the scrolling input system **200**. Further, a user can use a button pad **240** to command the robotic drive **24** to move or rotate an EMD without simultaneous use of a scrolling input system **200**. For example, the user can use one or both hands to manipulate the binary motion control **250, 252** of FIG. **24** or the scrolling motion controls **256** of FIG. **25** to control movement of multiple EMDs.

(111) In some embodiments, a scrolling input system can include alternative input control configurations. For example, FIG. **18** illustrates a scrolling input system **200** that uses two touch input controls **246** in place of the two scroll wheels **216, 226** of FIGS. **17** and **18**. A user may slide a finger along the touch input controls **246**, which may be configured as capacitive touch pads, in order to simulate a scrolling input control. In another embodiment, at least one of the scroll wheels **216, 226** may be omitted and binary buttons may be provided to control axial or rotational motion of the EMD. Some embodiments of a catheter-based procedure system may include at least one additional button pad and/or at least one additional scrolling input control, which may be different or the same as the illustrated embodiments. Further, although the scrolling input systems illustrated in FIGS. **15-19** are configured to be grasped by the user's right hand, it should be appreciated that similar scrolling input systems can include a body that is configured to be grasped by a user's left hand.

(112) FIG. **19** illustrates a perspective view of a translating input system **300**, as well as a button pad **240**, that can be used to control the robotic drive **24** based on linear movement of the translating input system **300**. The translating input system **300** includes a body **304** that is configured to slide along a guide rail **308**, and which can be held in one of the user's hands. The illustrated guide rail **308** constrains movement of the translating input system **300** to a single axis, thereby limiting its movement to a forward direction (away from the user) and a backwards direction (towards the user). Other embodiments may be constrained in different directions. For example, a translating input system may be configured to slide on a rail that constrains the translating input system to lateral (left and right) motion relative to the user. In another embodiment, the movement of a translating input system may be physically unconstrained so that it can be moved in any direction. An unconstrained translating input system can be configured to control axial and rotational motion of an EMD based on its movement along two perpendicular axes. Alternatively, an unconstrained translating input system can be configured to only measure movement along one axis, or the control computing system **34** may be configured to only interpret movement of the translating input system along a single axis.

(113) With continued reference to FIG. **19**, the translating input system **300** can be configured to instruct the robotic drive **24** to move or rotate an EMD based on its movement along the rail **308**. For example, the translating input system **300** can be configured to control axial movement of a selected EMD in a position control mode. When in a position control mode, the translating input system **300** can instruct the robotic drive **24** to move the EMD by a prescribed increment each time the translating input system **300** moves a nominal linear distance. The direction of the commanded EMD movement can be based on which direction the user moves the translating input system **300**. Thus, in some embodiments, a user can instruct the robotic drive **24** to move the EMD in the distal direction by moving the translating input system **300** in the forward direction, or in the proximal direction by moving the translating input system in the backwards direction.

(114) The range of linear the movement for the translating input system **300** may be constrained by various different boundaries, for example, a stop member defining an end of the rail **308**, the size of, or available space at, the control station, the physical reach of the user, or any other limiting factor or structure. In order to continue moving the translating input system **300** once a limit has been reached, translating input system **300** must be moved back towards the opposite end of its range of motion without actuating commanding movement of the EMD. In some embodiments, a user may be able to lift the translating input system **300** up off of the surface on which it slides. In such an embodiment, the user can lift up the translating input system, move it back towards the other end of its range of motion, and place it back down without instructing the robotic drive **24** to move the EMD.

(115) Additionally or alternatively, some embodiments of the translating input system **300** can include a continuous activation button **312** that the user must hold in order to control axial or rotational motion of the EMD. While the continuous activation button **312** is not being pressed by the user, the robotic drive **24** will not move the selected

EMD in response to movement of the translating input system **300**. In addition to helping to prevent incidental axial and rotational movement of the EMD, the continuous activation button **312** allows the user to return the translating input system **300** to its operable range by simply moving it while the button **312** is not pressed.

(116) Similar to the previously discussed scrolling input controls, the relationship between how far the user moves the translating input system **300** and how far the robotic drive **24** moves the EMD can be configurable or fixed. Some embodiments may include at least one scaling input on the translating input system **300** or the button pad that can be used to increase or decrease the how far the robotic drive **24** moves the EMD when the translating input system **300** is moved any given distance. Further, the control computing system **34** may adjust the ratio of commanded EMD motion to movement of the translating input system **300** based on which EMD and/or device module **32** is selected, or based on other system parameters.

(117) In some embodiments, the body **304** of the translating input system **300** may be similar to that of a scrolling input system illustrated in any of FIGS. **15-18**. A side surface **320** of the translating input system **300** can include a scroll wheel **316**, which can be in a similar or the same position as the secondary scroll wheel **226** in FIGS. **16** and **17**. The scroll wheel **316** can be configured as a motion control that may allow the user to control rotational motion of the EMD while axial movement is simultaneously controlled by sliding the translating input system **300**. Additionally or alternatively, some translating input systems can include at least one other input control positioned on the top surface **324** of the translating input system **300**, the side surface, or at any other location. Further, the translating input system **300** can be used in conjunction with a button pad **240** that includes multiple binary buttons **242** that may be manipulated by the user to adjust the motion commands produced using the translating input system **300**. For example, as with the button pad of FIG. **17**, the binary buttons **242** can be configured as selection controls for selecting which of the EMDs and/or device modules **32** will be controlled using the translating input system **300**.

(118) In some embodiments, the control station **26** can include a user interface configured to provide a user with feedback and information regarding the state of the catheter-based procedure system **10**. For example, the display **30** can provide a graphical user interface (GUI) that illustrates the positions and travel limits of the device modules **32** of a robotic drive **24**. FIGS. **20** and **21** illustrate embodiments of a GUI **400** configured to provide position information for a robotic drive **24** with three device modules **32**, which are denoted with the labels “C1,” “C2,” and “C3.” The three device modules **32** are each represented by an icon **404**, **406**, **408** that is arranged linearly within a boundary region **412** that represents the total length of the robotic drive **24**. To help the user to determine which icon **404**, **406**, **408** corresponds to each of the device modules **32**, each of the icons can include a text label **416** that denotes which device module **32** the icon **404**, **406**, **408** represents. Additionally, the icons **404**, **406**, **408** can be vividly colored or have different fill patterns to help a user to differentiate between each of the icons **404**, **406**, **408**. The position of each icon **404**, **406**, **408** corresponds to the position of the corresponding device module **32** on the robotic drive **24**. In the illustrated embodiments, the left side of the boundary region **412** represents the distal end of the robotic drive **24** while the right side represents the proximal end of the robotic drive **24**. In other embodiments, however, a GUI can be configured in other orientations.

(119) The GUI **400** can also include range bars **420**, **422**, **424** that illustrate the travel ranges of each of the device modules **32**. Each of the range bars **420**, **422**, **424** can have a color, fill pattern, or other design feature that corresponds to the design of the corresponding one of the icons **404**, **406**, **408**. For example, the range bars **420**, **422**, **424** can each be filled with a lighter shade of the color used for the corresponding icons **404**, **406**, **408**. To illustrate overlapping travel ranges, the range bars **420**, **422**, **424** can have different widths so that they appear to be layered on top of each other. For example, the range bar **420** of the first device module **32** (represented by the leftmost icon **404**) appears to be behind the range bar **422** of the second device module **32** (represented by the middle icon **406**).

(120) In the illustrated embodiments, the distal travel limit for each device module **32** is defined by the position of the device module **32** in front of it (in the distal direction), or by the distal end of the robotic drive **24**. Thus, the range bar **420** of the first device module **32** is attached to the left side of the boundary region **412**, while the range bars **422**, **424** of the second and third device modules **32** (represented by the middle icon **406** and rightmost icon **408**, respectively) are attached to the right side (proximal end) of the adjacent device module **32** in the distal direction. The proximal travel limit for each device module **32** is defined by a system-implemented limit (which may be defined by the user or by the system) as well as the position of the device module **32** behind it (in the proximal direction), or by a limit defined by the user or by the system. Thus, the right side of each of the range bars **420**, **422**, **424** (representing the proximal travel limits) is spaced apart from the left side by a distance that corresponds to the distance between the distal travel limit and the system-defined proximal travel limit. However, because the distal travel limit of the device modules **32** is defined by the proximal side of the adjacent device module **32**, each device module **32** can be moved inside of the travel range of the distally-adjacent device module **32**. In FIG. **20**, for example, the range bar **424** of the third device module **32** (which is represented by the rightmost icon **408**) overlaps with the range bar **422** of the second device module **32**.

(121) As illustrated in FIG. **21**, the maximum travel range of a device module (in this case, the range bar **424** of the third device module **32**) may be illustrated as extending outside of the boundary region **412**. Although the proximal travel limit of the third device module in this case is the proximal end of the robotic drive **24**, this may be useful in

order to show that the maximum travel range of the third device module **32** is longer than the distance between the proximal end of the second device module **32** and the proximal end of the robotic drive. In some embodiments, the GUI **400** can be configured to display additional graphics or text when a device module **32** reaches a travel limit. The additional graphics or text may indicate where the limit, what is defining the travel limit, and how the user can control the robotic drive **24** to continue moving the device module **32**. Further, the GUI **400** may highlight or otherwise point out the “contact point” between device modules **32** on the graphical illustration of their positions. In addition to displaying the proximal end of EMDs (as shown in FIG. **20** and FIG. **21**), in one embodiment the distal tip of EMDs can be displayed (not shown). This may be helpful to drive an EMD inside another EMD without use of fluoroscopy. Additionally, GUI **400** may communicate the device presence or absence, the cassette presence or absence, and notifications such as encoder mismatch, encoder issues, transducer issues, and any other notifications.

(122) While the illustrated GUI **400** is shown in a configuration for a catheter-based procedure system **10** with three device modules, other embodiments can be used with a catheter-based procedure system that includes more than three or fewer than three device modules. Depending on the configuration of the robotic drive **24** a GUI can change to display the appropriate number of device modules and their travel ranges. GUI **400** may also include numeric readout of device module position, which could be in reference to its absolute position along the full range of the system, or its remaining travel distance in either direction. Further, GUI **400** can be configured to indicate which of the device modules **32** have been selected by the user. For example, the icons **404**, **406**, **408** can be configured to switch between a selected state indicating that the user has selected that corresponding device module **32** using the input system **28**, and an unselected state indicating that the device module **32** has not been selected (and therefore will not be controlled using the motion controls).

(123) In some embodiments, a GUI may also provide a graphical representation of the input system **28** and the input controls on input system (for example, the buttons, knobs, joysticks, or any other input controls) using an input map (not illustrated). As the input controls are manipulated by the user, the button map can indicate which input controls are being activated. This may be useful so that the user may see which input controls are being manipulated without looking at the input system **28**. For example, the button map can include an array of icons corresponding to each or the input controls, which may be arranged on screen in a pattern that is similar to the physical arrangement of the input controls on the input system **28**. Each icon can be configured to light up and/or be animated when manipulated by the user. For binary input controls, corresponding icon on the button may indicate whether the input control is in an activated (pressed) or inactivated (unpressed) state. The icons corresponding to analog input controls may indicate a degree of manipulation.

(124) In some embodiments, a GUI can be configured to provide a graphical representation of different forces and/or torques that are acting on an EMD or a device module **32**. For example, FIG. **22** illustrates a linear gauge **430** configured to show a measured load, and FIG. **23** illustrates a rotary gauge **434** for illustrating a measured load. Each gauge **430**, **434** includes a pointer **438** configured to point to a position on the gauge **430**, **434** corresponding to the amount of force or torque acting on the EMD or device module **32**. Multiple different zones **440**, **442**, **444** can be marked along the gauges **430**, **434**, each corresponding to different load thresholds. The zones **440**, **442**, **444** can have different colors and/or fill patterns in order to help the user to differentiate between zones. In other embodiments, a GUI can include a load gauge in a different shape, size, or configuration. Further, a GUI can include a load gauge that does not include distinct zone designations. In another embodiment the graphical representations of measured loads are applied to and integrated with the robotic drive user interface shown in FIG. **20** and FIG. **21**.

(125) Along with a graphical representation, the GUI can include a numeric readout of the measured load. Further, the measured force or torque may be communicated by changes in color shade and/or intensity. A light can be configured to change between discrete colors and intensities as different load thresholds are passed, or it may be configured to gradually change or fade between different colors and intensities. For example, a numeric readout of a measured load can be configured to fade from a first color associated with a minimum or no measured load, to a second color associated with a maximum load limit. Additionally or alternatively, a load indicator light, which may be shown on the display **30**, the input system **28**, or any other location on the control station **26**, may change color based on the measured load. The value of the different load thresholds, including the maximum load limit, may be constant or they can be programmable and adjustable by user to accommodate different limits. This may be useful, for example, so that different users can set different limits based on their personal preference. Further, the control computing system **34** can be configured to prevent the user from exceeding the maximum load limit. Alternatively, the maximum load limit may be displayed on the display **30**, but the user may be able to override it.

(126) In addition to visual feedback, the control station **26** may be configured to provide physical feedback to the user. For example, feedback may be provided in the form of at least one of vibration, cogging, and a resistance or counteracting force. In some embodiments, vibration feedback may be used to provide various alerts. The vibrations may be provided at on preset, constant intensity, or the intensity may vary based on different levels of the alerts. For example, the intensity of vibrations may increase as a measured load approaches a maximum load limit. Alternatively, different intensities or vibration patterns may be used to differentiate between different alerts. Cogging feedback refers to the sensation of incremental bumps or clicks that give the user a sensation of a control mechanism traveling. In some

embodiments, this may be similar to the tactile feedback provided by a scrolling input control as it is rotated between positions created by its detents. Some systems, however, can use at least one different sensation for cogging feedback. The type and intensity of the cogging sensation may be fixed, or it may be adjustable by the user or by the system.

(127) Physical feedback may be by physical interactions between components that occur as the user manipulates the input system **28**, or it can be simulated using electromechanical devices. For example, a motor can be controlled to provide a tapping sensation as cogging feedback. When physical feedback is provided through an input system **28**, the feedback may be felt throughout the input system, it may be provided near (or appear to originate from) an area located close to specific input controls. For example, vibration feedback can be provided by a motor (or any other device) that is near an input control to indicate alerts associated with that input control.

(128) In some embodiments, input controls configured a motion controls be configured to provide physical feedback associated with a force or torque measured on the EMD or device module **32** being controlled by the input control. An analog trigger, a scroll wheel, a slider, or a translating input system may include a braking mechanism configured to resist manipulation by the user. In some embodiments the braking force may be provided by a passive system, such as a dampening system. Other embodiments can include a motor configured to apply a counteracting force to resist manipulation of the input control by the user. The strength of the counteracting force may be adjustable by the system or by the user and can vary based on a load measured by the system.

(129) Computer-executable instructions for controlling a catheter-based procedure system according to methods using any of the above-described components may be stored on a form of computer readable media. Computer readable media includes volatile and nonvolatile, removable, and non-removable media implemented in any method or technology for storage of information such as computer readable instructions, data structures, program modules or other data. Computer readable media includes, but is not limited to, random access memory (RAM), read-only memory (ROM), electrically erasable programmable ROM (EEPROM), flash memory or other memory technology, compact disk ROM (CD-ROM), digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired instructions and which may be accessed by system **10** (shown in FIG. **1**), including by internet or other computer network form of access.

(130) The patentable scope herein is defined by the claims, and may include other examples that occur to those in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims. The order and sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments.

(131) Many other changes and modifications may be made to the embodiments described herein without departing from the spirit thereof. The scope of these and other changes will become apparent from the appended claims.

Claims

1. An input system for controlling a catheter-based procedure system that includes a robotic drive, the input system comprising: a body comprising a top surface and a front surface and; a first control disposed on the front surface and configured to instruct the robotic drive to axially move a first elongated medical device in response to manipulation of the first control by a user; and a second control disposed on the top surface and configured to instruct the robotic drive to rotate the first elongated medical device in response to manipulation of the second control by the user, wherein the first control and the second control are disposed to allow manipulation of the first control by an index finger of the user and manipulation of the second control by a thumb of the user.
2. An input system according to claim 1, further comprising a third control coupled to the body, wherein the first control and the second control are configured to instruct the robotic drive only if the third control is manipulated by the user.
3. An input system according to claim 2, wherein the first control is configured to instruct the robotic drive to axially move the first elongated medical device at a constant speed in response to manipulation of the first control by a user, and wherein the second control is configured to instruct the robotic drive to rotate the first elongated medical device by a discrete angle in response to manipulation of the second control by the user.
4. An input system according to claim 1, further comprising a third control coupled to the body and configured to map the first control and the second control to a second elongated medical device, wherein mapping the first control and the second control to the second elongated medical device results in configuration of the first control to instruct the robotic drive to axially move the second elongated medical device in response to manipulation of the first control by the user, and configuration of the second control to instruct the robotic drive to rotate the second elongated medical device in response to manipulation of the second control by the user.
5. An input system according to claim 4, further comprising a fourth control coupled to the body, wherein the first control and the second control are configured to instruct the robotic drive only if the fourth control is manipulated by the user.

6. An input system according to claim 5, further comprising a fifth control coupled to the body and configured to map the third control and the fourth control to a third elongated medical device, wherein mapping the third control and the fourth control to the third elongated medical device configures the third control to instruct the robotic drive to axially move the third elongated medical device in response to manipulation of the third control by the user and configures the fourth control to instruct the robotic drive to rotate the third elongated medical device in response to manipulation of the fourth control by the user.
7. An input system according to claim 1, further comprising a third control coupled to the body, wherein the first control and the second control are configured to instruct the robotic drive only if the third control is manipulated by the user.
8. An input system according to claim 1, wherein the first control is configured to instruct the robotic drive to axially move the first elongated medical device a continuous speed in response to manipulation of the first control by a user, and wherein the second control is configured to instruct the robotic drive to rotate the first elongated medical device by a discrete angle in response to manipulation of the second control by the user.
9. An input system according to claim 1, further comprising: a third control disposed on the front surface and configured to instruct the robotic drive to axially move a second elongated medical device in response to manipulation of the third control by a second index finger of a second hand of the user.
10. An input system according to claim 9, further comprising: a fourth control coupled to the body and configured to map the first control and the second control to a selected one of two or more elongated medical devices; and a fifth control coupled to the body and configured to map the third control to a second selected one of two or more elongated medical devices.
11. An input system according to claim 10, wherein the fourth control is configured to map the first control and the second control to the selected one of two or more elongated medical devices only while the fourth control is manipulated by the user, and wherein the fifth control is configured to map the third control to the second selected one of two or more elongated medical devices only while the fifth control is manipulated by the user.
12. A method for an input system for controlling a catheter-based procedure system that includes a robotic drive, the method comprising: receiving a first manipulation by an index finger of a first hand of a user of a first control coupled to a front surface of a body of the input system; receiving a second manipulation by a thumb of the first hand of the user of a second control coupled to a top surface of the body of the input system; responsive to the first manipulation, instructing the robotic drive to axially move a first elongated medical device; and responsive to the second manipulation, instructing the robotic drive to rotate the first elongated medical device, wherein the first manipulation and the second manipulation occur simultaneously.
13. A method for an input system according to claim 12, further comprising: receiving a third manipulation of a third control coupled to the body, wherein the robotic drive is instructed only if the third manipulation is received simultaneously with the first manipulation and the second manipulation.
14. A method for an input system according to claim 13, wherein the first control is configured to instruct the robotic drive to axially move the first elongated medical device a continuous speed responsive to manipulation of the first control by a user, and wherein the second control is configured to instruct the robotic drive to rotate the first elongated medical device by a discrete angle responsive to manipulation of the second control by the user.
15. A method for an input system according to claim 12, further comprising: receiving a third manipulation of a third control coupled to the body; and responsive to the third manipulation, mapping the first control and the second control to a second elongated medical device.
16. A method for an input system according to claim 15, further comprising receiving a fourth manipulation of a fourth control coupled to the body, wherein the robotic drive is instructed only if the fourth manipulation is received simultaneously with the first manipulation and the second manipulation.
17. A method for an input system according to claim 15, wherein the first control and the second control are mapped to the second elongated medical device only while the third control is manipulated by the user.
18. A method for an input system according to claim 12, further comprising: receiving a third manipulation by an index finger of a second hand of the user of a third control coupled to the front surface of the input system; receiving a fourth manipulation by a thumb of the second hand of the user of a fourth control coupled to the top surface of the input system; responsive to the third manipulation, instructing the robotic drive to axially move a second elongated medical device; and responsive to the fourth manipulation, instructing the robotic drive to rotate the second elongated medical device, wherein the third manipulation and the fourth manipulation occur simultaneously.
19. A method for an input system according to claim 12, wherein the first control is configured to instruct the robotic drive to axially move the first elongated medical device a continuous speed responsive to manipulation of the first control by a user, and wherein the second control is configured to instruct the robotic drive to rotate the first elongated medical device by a discrete angle responsive to manipulation of the second control by the user.
20. A method for an input system according to claim 12, further comprising: receiving a third manipulation of a third control coupled to the front surface of the body; and responsive to the third manipulation, instructing the robotic drive to axially move a second elongated medical device.

